



Submersible Pump in Discharge Tube

AmaCan D

50 Hz

Type Series Booklet



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Type Series Booklet AmaCan D

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Water applications: Water Transport / Waste Water Transport, Flood Protection

Submersible Pumps in Discharge Tubes

AmaCan D



Main applications

- Irrigation pumping stations
- Drainage pumping stations
- Stormwater pumping stations
- Raw water pumps and clean water pumps in waterworks and waste water treatment plants
- Cooling water pumps in power stations and industry
- Industrial water supply
- Water pollution control
- Flood control
- Aquaculture

Fluids handled

- Waste water
- Sludge
- Surface water
- Stormwater
- Grey water
- Seawater
- Brackish water

Operating data

Table 1: Operating properties

Characteristic		Value
Flow rate	Q [l/s]	≤ 2250
	Q [m³/h]	≤ 8100
Head	H [m]	≤ 29
Motor rating	P ₂ [kW]	≤ 340
Fluid temperature	T [°C]	≤ +40

Designation

Example: AmaCan DB3 600-420/1304UTG1 IE3

Table 2: Designation key

Code	Description
AmaCan	Type series
D	Impeller type
D	Open mixed-flow multi-vane impeller
B	Impeller variant
A	
B	
3	Number of vanes
2	
3	
600	Nominal diameter of the discharge tube [mm]
420	Nominal impeller diameter [mm]
130	Motor size
4	Number of motor poles
UT	Motor version
UT, UV	Non-explosion-proof
XT, XV	Explosion-proof Ex db h IIB T3 Gb
G1	Material variant
G1	Grey cast iron, standard material variant
G2	Cast iron, casing wear ring made of stainless steel
G3	Grey cast iron with Zn anodes, casing wear ring made of stainless steel, shaft made of stainless steel 1.4057
IE3	Motor efficiency classification ¹⁾
_2)	No efficiency classification
IE3	Premium Efficiency

Design details

Design

- Fully floodable submersible pump in discharge tube (submersible motor pump)
- Not self-priming
- Close-coupled design
- Single-stage
- Vertical installation

¹ The IEC 60034-30 standard is not binding for submersible motor pumps. Efficiencies are calculated/determined by analogy with the measurement method specified in IEC 60034-2. The marking is used for submersible motors that achieve efficiency levels similar to those of standardised motors acc. to the IEC 60034-30 standard.

² Blank

Drive

- Three-phase asynchronous squirrel-cage motor
- Enclosure: IP68 to EN/IEC 60529
- Type of protection Ex db IIB (applies to explosion-proof pump sets only)

Shaft seal

- Two bi-directional mechanical seals in tandem arrangement, with liquid reservoir
- Leakage chamber

Impeller type

- Open mixed-flow multi-vane impeller

Bearings

- Grease-lubricated rolling element bearings

Monitoring equipment

Various sensor packages are available for pump (set) monitoring:

- Basic**
 - Motor temperature monitoring (PTC thermistor)

- Leakage sensor in the motor space
- Mechanical seal leakage sensor (float switch)
- Bearing temperature monitoring, pump end (Pt100)

Basic³⁾

- Motor temperature monitoring (PTC thermistor)
- Leakage sensor in the motor space
- Mechanical seal leakage sensor (float switch)
- Bearing temperature monitoring, drive end (Pt100)
- Bearing temperature monitoring, pump end (Pt100)

Premium

- Motor temperature monitoring (PTC thermistor)
- Leakage sensor in the motor space
- Mechanical seal leakage sensor (float switch)
- Bearing temperature monitoring, drive end (Pt100)
- Bearing temperature monitoring, pump end (Pt100)
- Vibration monitoring
- Motor temperature measurement (Pt100)

The sensor information can optionally be evaluated via AmaControl using the live diagnosis function:

- Directly on site with the KSB INTspecter app
- In additional external systems via ModBus

Materials

Table 3: Overview of materials depending on material variant (EN and ASTM)

Part No.	Description	Material variant		
		G1	G2	G3 ⁴⁾ (seawater variant)
112	Pump bowl	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
138	Bellmouth	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
230	Impeller	EN: 1.4593/1.4517/ASTM: A995 Gr. 1B		
350 / 330	Bearing housing / bearing bracket	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
360	Bearing cover	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
412	O-ring	EN/ASTM: NBR ⁵⁾ EN: (Viton-FPM) ⁶⁾ /ASTM: (Viton-FKM ⁶⁾)		
433	Mechanical seal (pump end)	SiC/SiC		
	Mechanical seal (drive end)	Carbon/SiC		
502	Casing wear ring with a groove for flushing	EN: EN-GJL-250/ ASTM: A48 Cl. 35B	EN: 1.4593/1.4517/ASTM: A995 Gr. 1B	
571	Bail	EN: EN-GJS-400-15/S235JR/ASTM: A 536: 60–40–18/A284B		
811	Motor housing	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
812	Motor housing cover	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
818	Shaft (rotor)	EN: 1.4021/20Cr13/ ASTM: A 276 Type 420		EN: 1.4057/ ASTM: A 276 Type 431
82-5	Adapter	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
834	Cable gland	-		
	Cable gland housing	EN: EN-GJL-250/ASTM: A48 Cl. 35B		
Var.	Bolts/screws	Stainless steel		
99-16	Anode	-	-	Zn
Other materials on request.				

3 Exception: The Basic+ sensor package is not available in combination with the UV/XV motors.

4 Pump set with cathodic protection (anodes to be checked every 6 to 12 months) and top coat of 250 µm

5 Nitrile rubber (Perbunan)

6 FPM/FKM fluorocarbon rubber variant available as an option against a surcharge

Description of materials

Duplex stainless steel (1.4517 or technically equivalent material)

This type of carbon steel is resistant to cavitation, has excellent strength values and is used for high circumferential speeds. An excellent resistance to pitting corrosion makes ferritic-austenitic stainless carbon steel a popular choice for pumping acidic waste water with a high chloride content as well as seawater and brackish water. Thanks to its good chemical resistance, e.g. against waste water containing phosphorous and sulphuric acid, this material is used in a wide range of applications in the chemical industry and process engineering. Pumps made of duplex stainless steel have a very long service life, even when handling brines, chemical waste water (pH 1 - 12), grey water and landfill leachate.

Coating and preservation

Paint

- **Surface treatment:** SA 2 1/2 (SIS 055900) AN 1865
- **Primer:** primer coat on unfinished casting
- **Top coat:** environmentally friendly KSB standard coating (RAL 5002)

Special coating

- Available on request (extra charge and a longer delivery period apply).

Product benefits

- The pump's own weight ensures self-centring seating in the discharge tube, and an O-ring seals it; quick to install or remove.
- Non-clogging low-maintenance design with large free passages reduces clogging risk and, consequently, maintenance work.
- The slim motor keeps discharge tube flow losses down.
- High-efficiency motors and variable hydraulic systems for optimum hydraulic efficiency and energy efficiency
- High reliability thanks to bearing temperature monitoring, vibration sensor, thermal motor protection, leakage sensors in the motor space and connection space as well as leakage monitoring of the mechanical seal system.
- Low-vibration hydraulic system; inlet ribs and optimised bellmouth for vortex-free inflow.
- Absolutely water-tight resin-sealed cable entries prevent any water from entering the motor – even in the event of a damaged cable.
- Highest operating reliability by casing wear ring with groove for flushing
- IloT⁷⁾-capable: Internal sensor module and AmaControl analysing device can be used.

Acceptance tests and warranty

Functional test

- Every pump undergoes functional testing to KSB standard ZN 56525.
- Hydraulic values are guaranteed to EN ISO 9906 Class 2B.

Acceptance inspections/tests

- Acceptance test to ISO/DIN or comparable standards available against a surcharge.
- Acceptance inspections/tests to Hydraulic Institute on request.

Warranty

- Quality is assured by means of an audited and certified quality assurance system to DIN EN ISO 9001.

Selection information

Information for pump selection

The guaranteed point of submersible pumps in discharge tubes is measured at a head 0.5 m above the motor (DIN 1184). The documented characteristic curves refer to this data. This must be taken into account when calculating system losses. The indicated heads and performance data apply to pumped fluids with a density $\rho = 1 \text{ kg/dm}^3$ and a kinematic viscosity ν of up to $20 \text{ mm}^2/\text{s}$.

- Adjust the power input to the density of the fluid handled:
 $P_2 (\text{required}) = \rho [\text{kg/dm}^3] (\text{fluid handled}) \times P_2 (\text{documented})$
- Select the operating point with the largest power input within an operating range. Select a motor size providing a power reserve to compensate the tolerances in the system characteristic / pump characteristic.

Table 4: Recommended motor power reserve⁸⁾

P_2 [kW]	Reserve	
	Mains operation	With frequency inverter
≤ 30	10 %	15 %
> 30	5 %	10 %

Intake chamber

Determine the minimum water level t_{imin} (diagram in general arrangement drawing):

The minimum water level t_{imin} is the water level required in the pump's suction chamber to ensure:

- that there is a sufficient liquid cover above the hydraulic system (propeller) (shown in diagram depending on pump size)
- that the pump does not draw in air-entraining vortices (shown in diagram depending on flow rate)
- that there is no cavitation in the hydraulic system (check against the $\text{NPSH}_{\text{required}}$ value indicated in the technical literature). The following conditions must be met:
 - $\text{NPSH}_{\text{available}} > \text{NPSH}_{\text{required}} + \text{safety allowance}$
 - $\text{NPSH}_{\text{available}} = 10.0 + (t_1 - t_3 - h_7/2)$
 - Safety allowance:
up to $Q_{\text{opt}} \Rightarrow 0.5 \text{ m}$
larger than $Q_{\text{opt}} \Rightarrow 1.0 \text{ m}$

Head (H)

The total pump head is composed as follows:

$$H = H_{\text{geo}} + \Delta H_v$$

H_{geo} (static head)

- Without discharge elbow: difference between the suction-side water level and the overflow edge
- With discharge elbow: difference between suction-side and discharge-side water level

⁷ IloT = Industrial Internet of Things

⁸ If larger power reserves are stipulated by local regulations, these larger reserves must be provided.

ΔH_v (losses in the system)

- Starting 0.5 m downstream of the pump: e.g. pipe friction, elbow, swing check valve, etc.

Inlet losses, riser losses and elbow losses

Losses are caused by the inlet, riser and elbow (and/or free discharge).

- Losses in the riser up to the indicated reference level (0.5 m above the motor) are taken into account in the documented characteristic curves.
- Inlet and elbow losses are system losses. These losses must be taken into account for selection.
- For information on structural requirements, pump installation and pump sump design please refer to the KSB know-how brochure "Planning Information: Amacan Submersible Pumps in Discharge Tubes" 1581.025.

Overview of product features / selection tables

Table of fluids handled

The table below for your guidance is based on KSB's long-standing experience. The data are standard values and are not to be considered as generally binding recommendations. More detailed advice is available from KSB. Make use of our laboratory's expertise when selecting materials.

Table 5: Selection aid for material variants and fluid temperature per fluid

Fluid handled ⁹⁾	Max. permissible fluid temperature	Material variant	Screen	Comments, further recommendations
	[°C]			
Waste water				
▪ Industrial, corrosive, non-abrasive, slightly acidic; pH value ≥ 6	40	G2	✓	Two-component high solid epoxy finish coat (RAL 5002) 250 µm required
▪ Industrial, corrosive, non-abrasive, with lacquer/paint/varnish suspensions	40	G2	-	Lacquer/paint/varnish suspension = free of solvents
▪ Industrial, corrosive, non-abrasive fluids containing faeces	40	G2	✓	-
▪ Industrial, corrosive, non-abrasive fluids not containing faeces	40	G2	✓	-
▪ Municipal, cleaned	40	G1	✓	-
Suspended solids, water/sand mixture	40	G2	-	Up to 200 mg/l
Sludge	40	G1	-	Up to 2 % dry solids content
Water, seawater and brackish water	25 ¹⁰⁾	G3	-	Anodes ¹¹⁾ and 2-component high solid epoxy finish coat (RAL 5002) 250 µm required
Water, cooling water	40	G1	-	-
Water, surface water				
▪ River water	40	G1	✓	-
▪ No details specified	40	G1	✓	-
▪ Lake water, fresh water	40	G1	✓	-
▪ Lake water, dam water	40	G1	-	-
Water, rainwater				
▪ With strainer	40	G1	-	-
▪ Without strainer	40	G1	✓	-
Water, raw water	40	G1	-	-
Water, contaminated water				
▪ Slightly contaminated water	40	G1	-	-
▪ Combined sewage, with strainer	40	G1	-	-
▪ Combined sewage, without strainer	40	G1	✓	-
▪ Combined sewage containing faeces	40	G1	✓	-
▪ Combined sewage not containing faeces	40	G1	✓	-
Water, clean water	40	G1	-	-

Table 6: Symbols key

Symbol	Description
✓	Required
-	Not required

⁹ Fluids to be handled which are not listed in this table usually require higher-grade materials. Consult us for details.

¹⁰ Consult us for t > 25 °C (stainless steel variant).

¹¹ Efficiency reduced by 2 % to 3 %; anode to be checked every 6 to 12 months

Table 7: Space between screen bars

Size	Required space between screen bars
	[mm]
600-390	30
600-420	40
700-390	30
700-420	40
700-460	40
800-460	40
800-580	60
900-630	80
900-650	80
1000-650	80

Overview of product features

Table 8: Material variants (G1, G2, G3)

Feature	Motor version			
	UVG/XVG	UTG/XTG		
Motor size				
4-pole	35 4 ... 60 4	70 4 ... 130 4	150 4 ... 215 4	275 4 ... 300 4
6-pole	22 6 ... 30 6	47 6 ... 120 6	155 6 ... 205 6	250 6 ... 340 6
8-pole	–	40 8 ... 95 8	120 8 ... 160 8	205 8 ... 290 8
10-pole	–	–	–	–
12-pole	–	–	–	–
Explosion protection				
Version U...	Non-explosion-proof			
Version X...	Ⓔ II2G Ex db h IIB T3 Gb			
Motor				
Starting method	DOL	DOL or star-delta (690 V only DOL)		
Voltage	400 V ¹²⁾			
Cooling	Cooled by surrounding fluid			
Immersion depth	30 m max.			
Power cable				
Type	"Overview of power cables", (⇒ Table 9)			
Length	10 m ¹³⁾			
Cable entry	Absolutely water-tight			
Sealing elements				
Elastomer seals	Nitrile butadiene rubber NBR ¹⁴⁾			
Shaft seal	Bellows-type mechanical seal ¹⁵⁾			
Monitoring equipment				
Sensor package	Basic/Basic+/Premium (⇒ Page 4)			
Coating	Environmentally friendly KSB standard coating, colour RAL 5002 ¹⁶⁾			
Installation	BU, BUS, BG, CU, CUS, CG, DU, DUS, DG			
Maximum fluid temperature				
Material variant G1	40 °C			
Material variant G2	40 °C			
Material variant G3	25 °C			
Tests/inspections				
Hydraulic system	KSB standard (ZN 56525) ¹⁷⁾			
General	KSB standard (ZN 56525) ¹⁷⁾			

Table 9: Power cable overview

Feature	Power cable and sensor cable		Sensor cable
	S1BN8-F	S07RC4N8-F	S05BN8-F
Design	Standard	Optional	Optional
Rated voltage	1000 V	750 V	–
EMC screening	–	✓	✓
Insulation material	EPR ¹⁸⁾	EPR ¹⁸⁾	Elastomer mixture
Maximum continuous temperature of insulation	90 °C	90 °C	90 °C
For permanent immersion in waste water to DIN VDE 0282-16/HD22.16	✓	✓	✓

12 Optional: 380 V, 415 V, 500 V, 690 V

13 Optional: up to 40 m

14 Optional: Viton = fluorocarbon rubber FPM

15 Optional: mechanical seal with covered spring

16 Optional: 250 µm

17 Optionally to ISO 9906/1/2/A

18 EPR = ethylene propylene rubber

Related documents

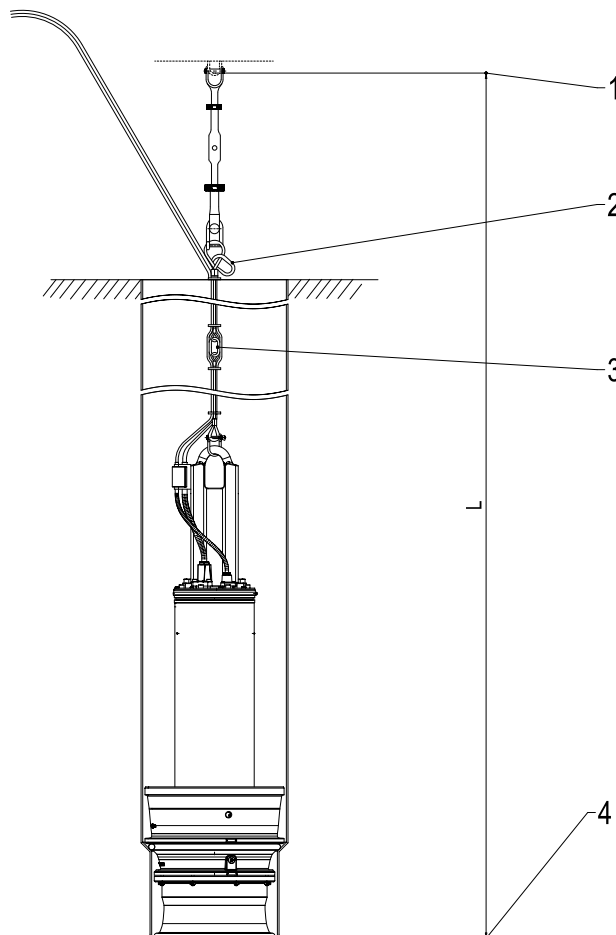
- General Arrangement Drawings 1581.39
- Planning Information 1581.025

Purchase order specifications

- Pump designation
- Flow rate Q , head H_{total}
- Type of fluid handled and fluid temperature
- Voltage, frequency, starting method, cable length
- Quantity and language of operating manuals
- Required accessories
 - For discharge tube drawings and discharge tubes, a drawing of the building indicating all required elevations and the type of installation is required.
 - For flow-straightening vanes indicate the type of installation and design (with or without suction umbrella).

- For a support rope indicate dimension "L", the number of additional lifting rings (depending on the lifting height of the lifting equipment) as well as the elevations and type of installation.

Always define dimension "L" when ordering a support rope to allow the correct length to be determined. The lifting height of the crane must be taken into account when ordering a support rope. This determines the number of lifting rings required for installing the pump in or removing it from the discharge tube.

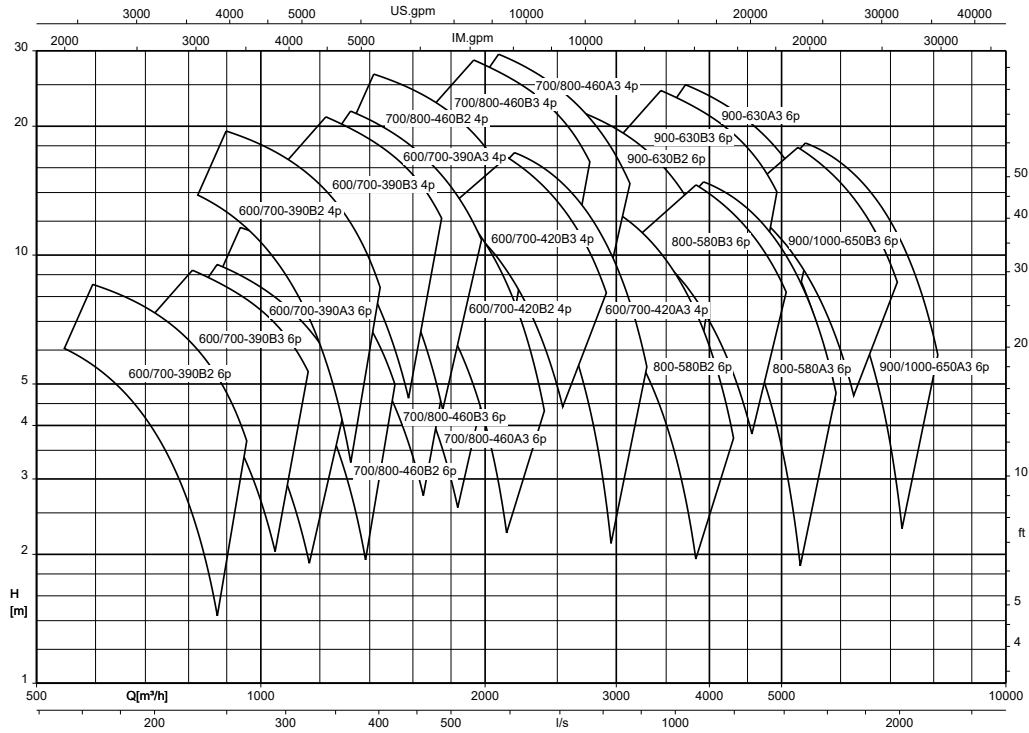


1	Suspension arrangement attached to cross-beam (BU/BUS/BG) or cover (CU/CUS/CG/DU/DUS/DG)
2	Lifting ring (standard, included in the scope of supply)
3	Optional (intermediate) lifting ring(s)
4	Lower edge of discharge tube = lower edge of pump

The support rope is an accessory and can be supplied with additional lifting rings and a support spacer as an option. The standard design is supplied without intermediate lifting ring(s).

Selection chart

AmaCan D 50 Hz, n = 1450 / 960 rpm (diameter-based selection chart)



Characteristic curves

$n = 1450 \text{ rpm}$

AmaCan DA3 600-390, $n = 1450 \text{ rpm}$

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed

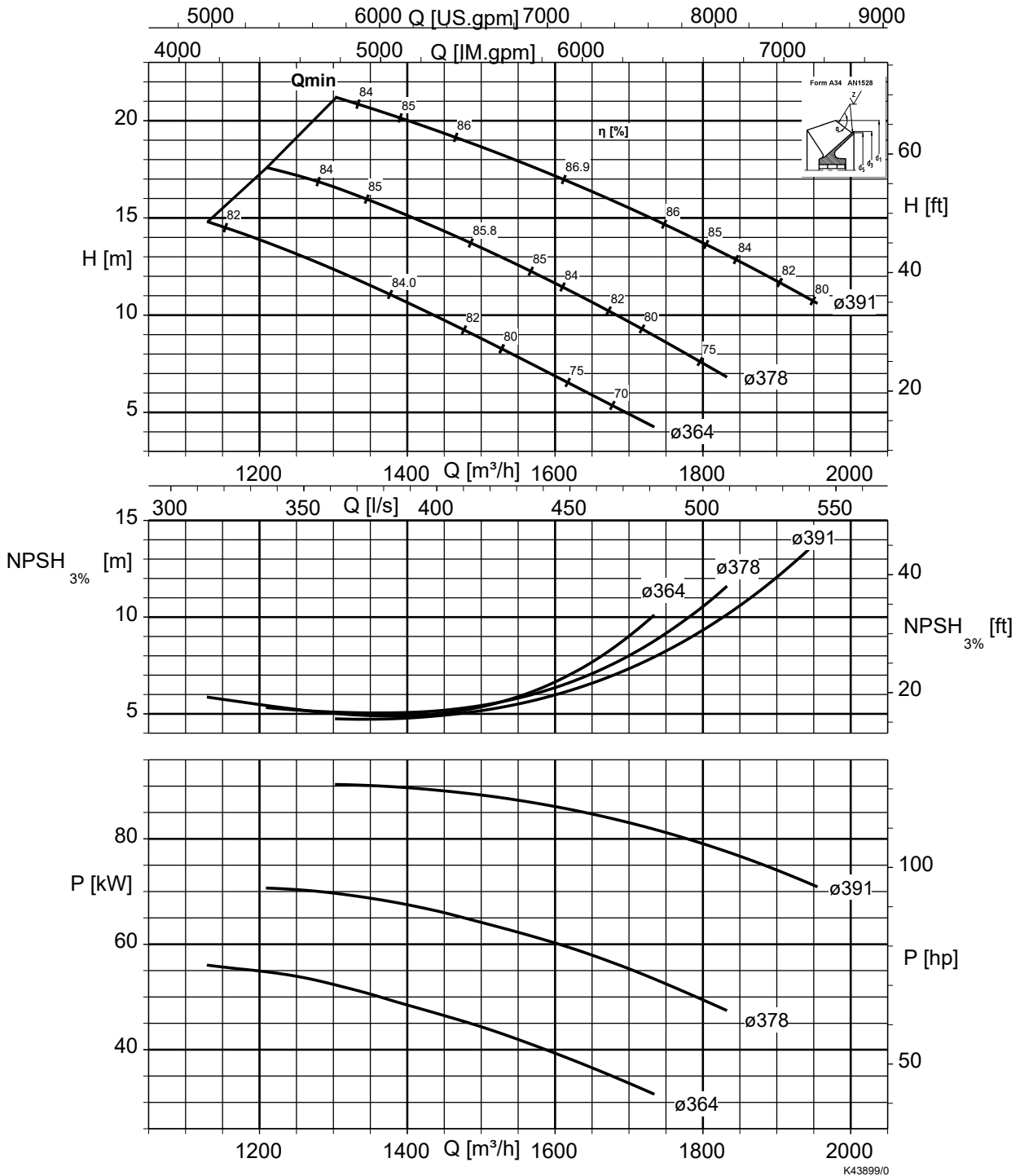


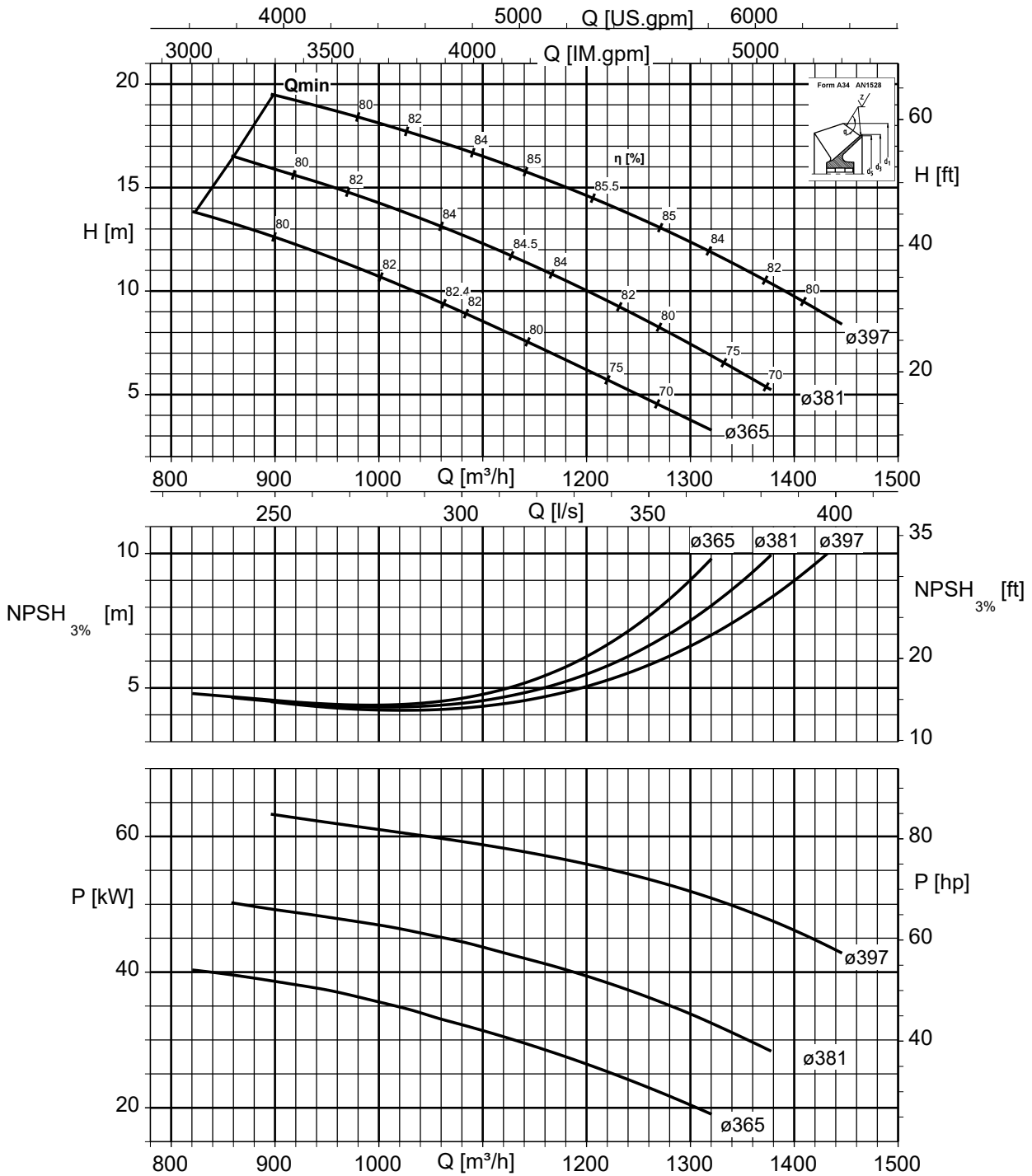
Table 10: Motor data

Size	Motor	P2 [kW]		J
		¹⁹⁾	IE3	[kgm²]
600-390	554 UV/XV	55	-	0,77
600-390	604 UV/XV	60	-	0,79
600-390	704 UT/XT	70	-	0,92
600-390	904 UT/XT	90	-	1,05
600-390	1054UT/XT	105	55	1,13
600-390	1304UT/XT	-	75	1,21

19 - = No efficiency classification

AmaCan DB2 600-390, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 65 mm

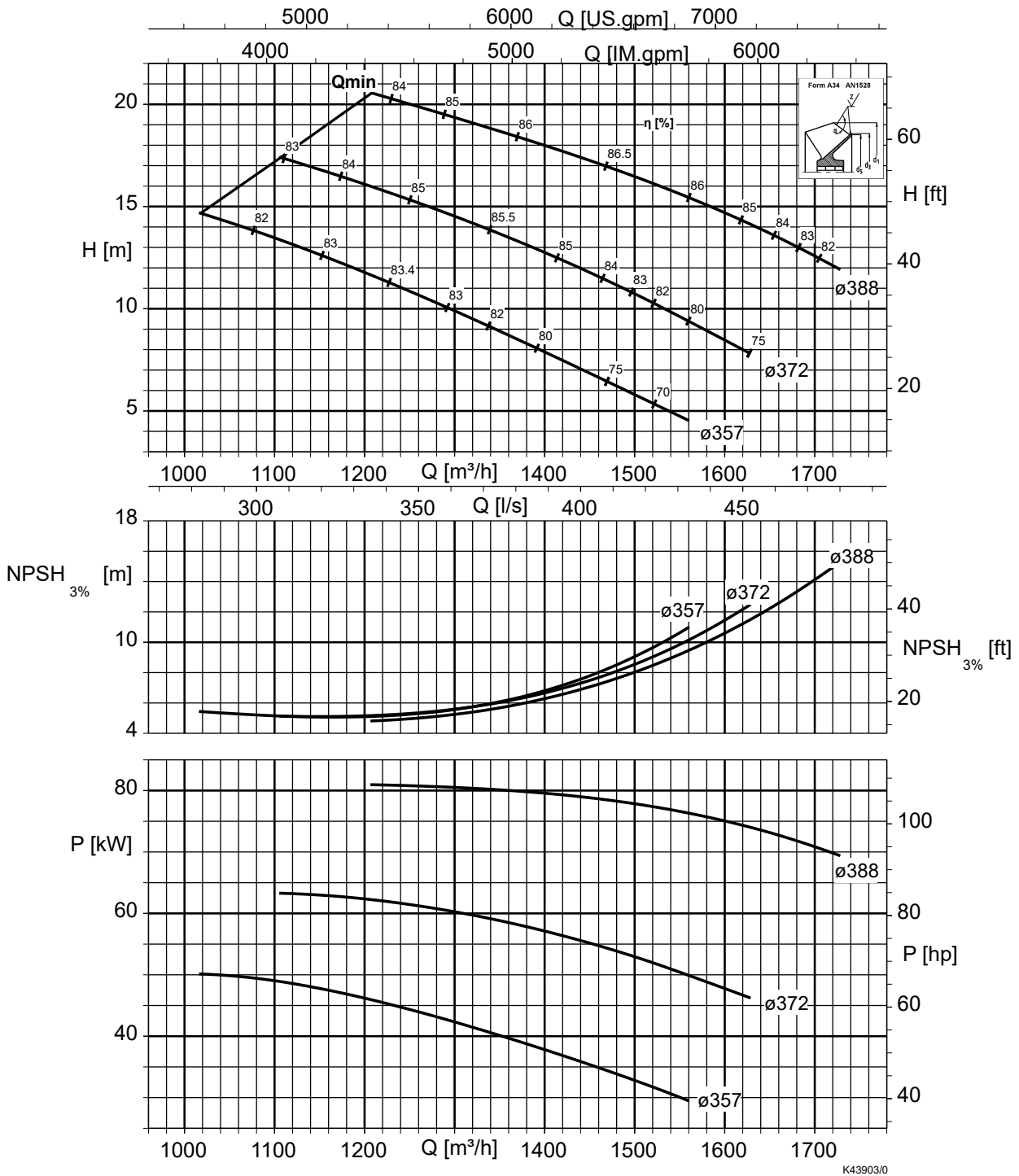
Table 11: Motor data

Size	Motor	P2 [kW]		J
		₋₂₀₎	IE3	[kgm²]
600-390	354 UV/XV	35	-	0,63
600-390	454 UV/XV	45	-	0,66
600-390	554 UV/XV	55	-	0,68
600-390	604 UV/XV	60	-	0,70
600-390	704 UT/XT	70	-	0,83
600-390	804 UT/XT	-	37	0,87
600-390	904 UT/XT	90	45	0,97
600-390	1054UT/XT	-	55	1,05
600-390	1304UT/XT	-	75	1,13

20 - = No efficiency classification

AmaCan DB3 600-390, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 65 mm

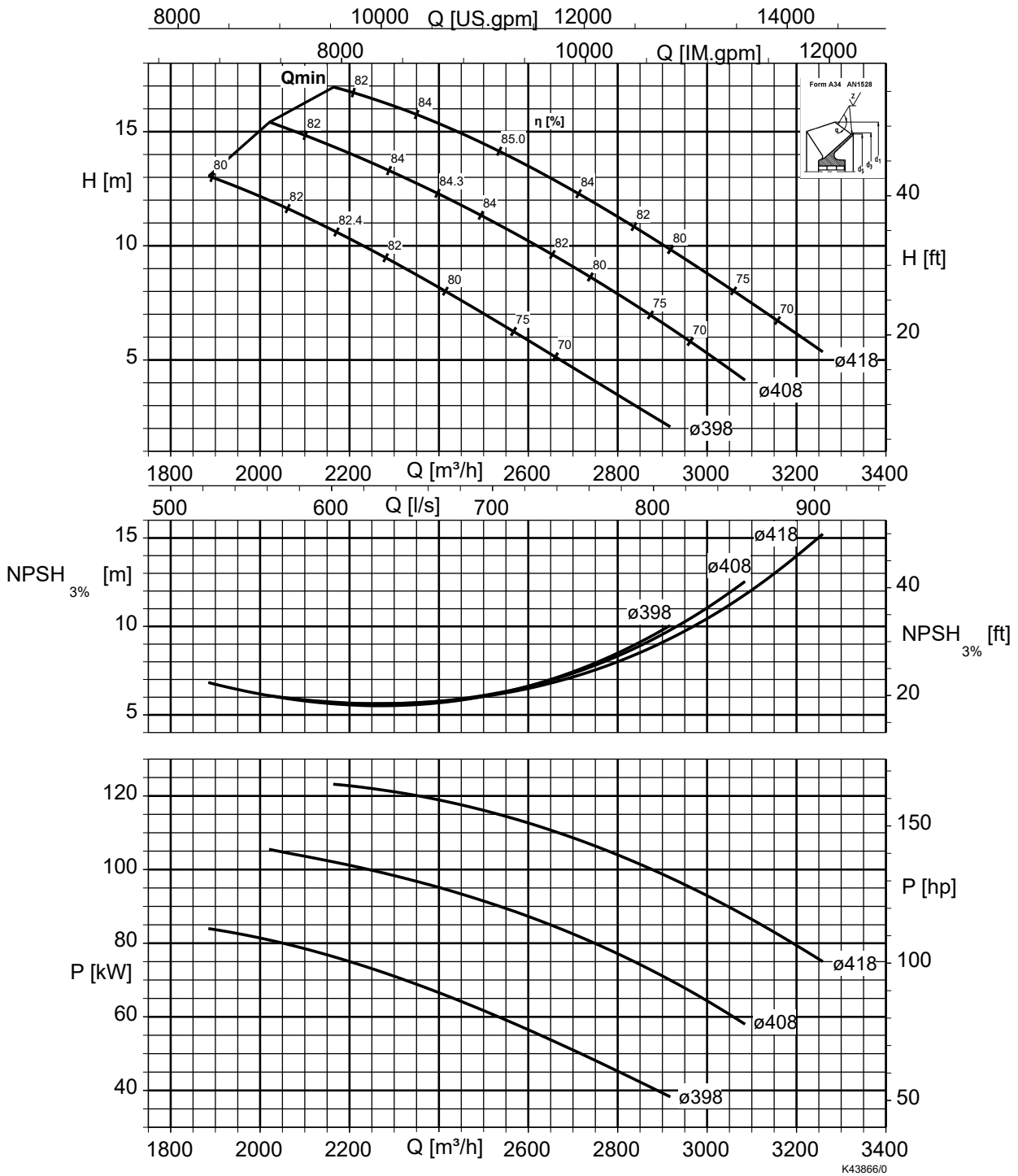
Table 12: Motor data

Size	Motor	P2 [kW]		J
		21)	IE3	[kgm²]
600-390	554 UV/XV	55	-	0,76
600-390	604 UV/XV	60	-	0,78
600-390	704 UT/XT	70	-	0,87
600-390	904 UT/XT	90	-	1,01
600-390	1054UT/XT	105	55	1,05
600-390	1304UT/XT	-	75	1,17

21 - = No efficiency classification

AmaCan DA3 600-420, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

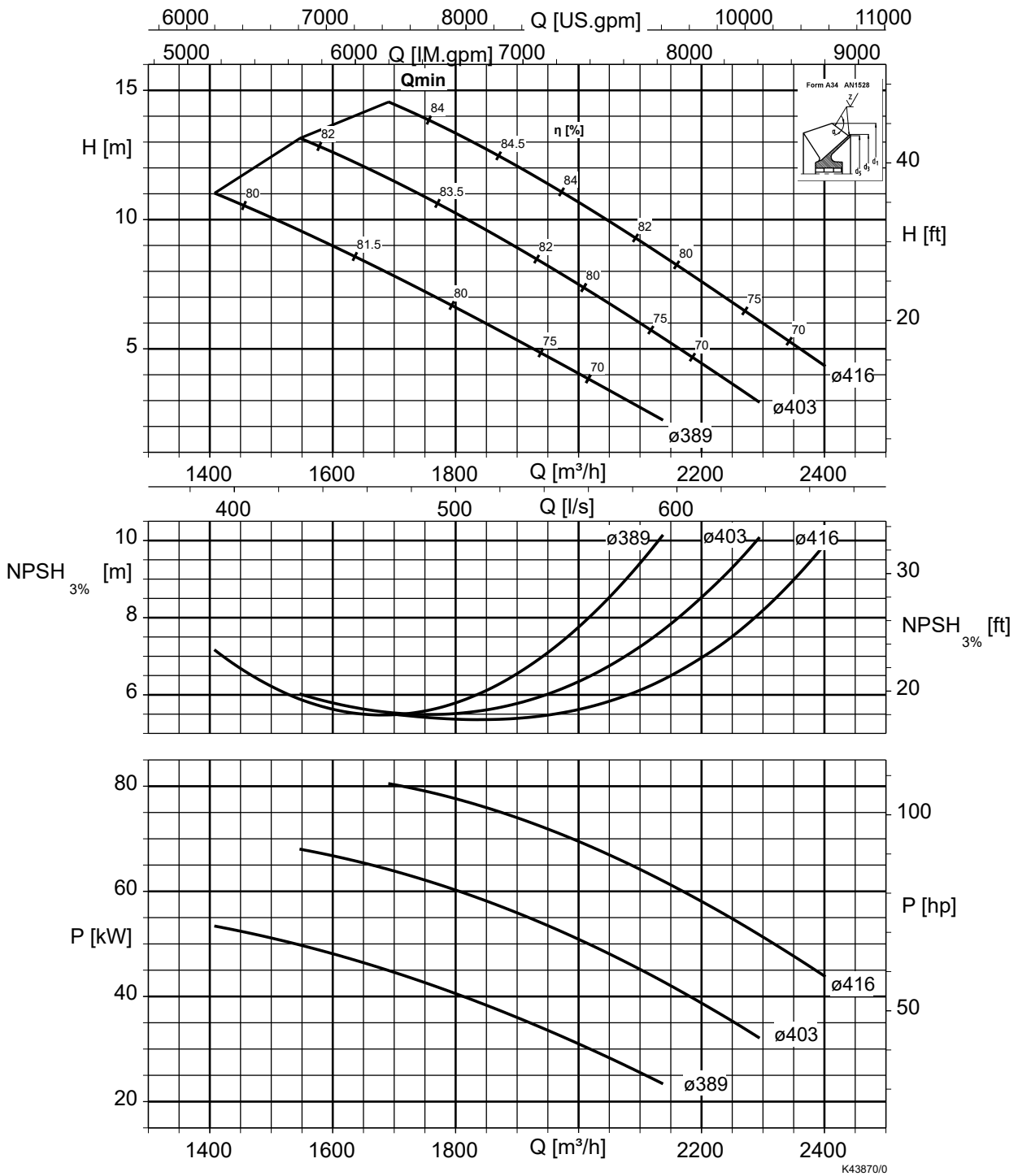
Table 13: Motor data

Size	Motor	P2 [kW]		J
		₋₂₂₎	IE3	[kgm ²]
600-420	904 UT/XT	90	-	1,13
600-420	1054UT/XT	105	-	1,21
600-420	1304UT/XT	130	-	1,29

22 - = No efficiency classification

AmaCan DB2 600-420, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

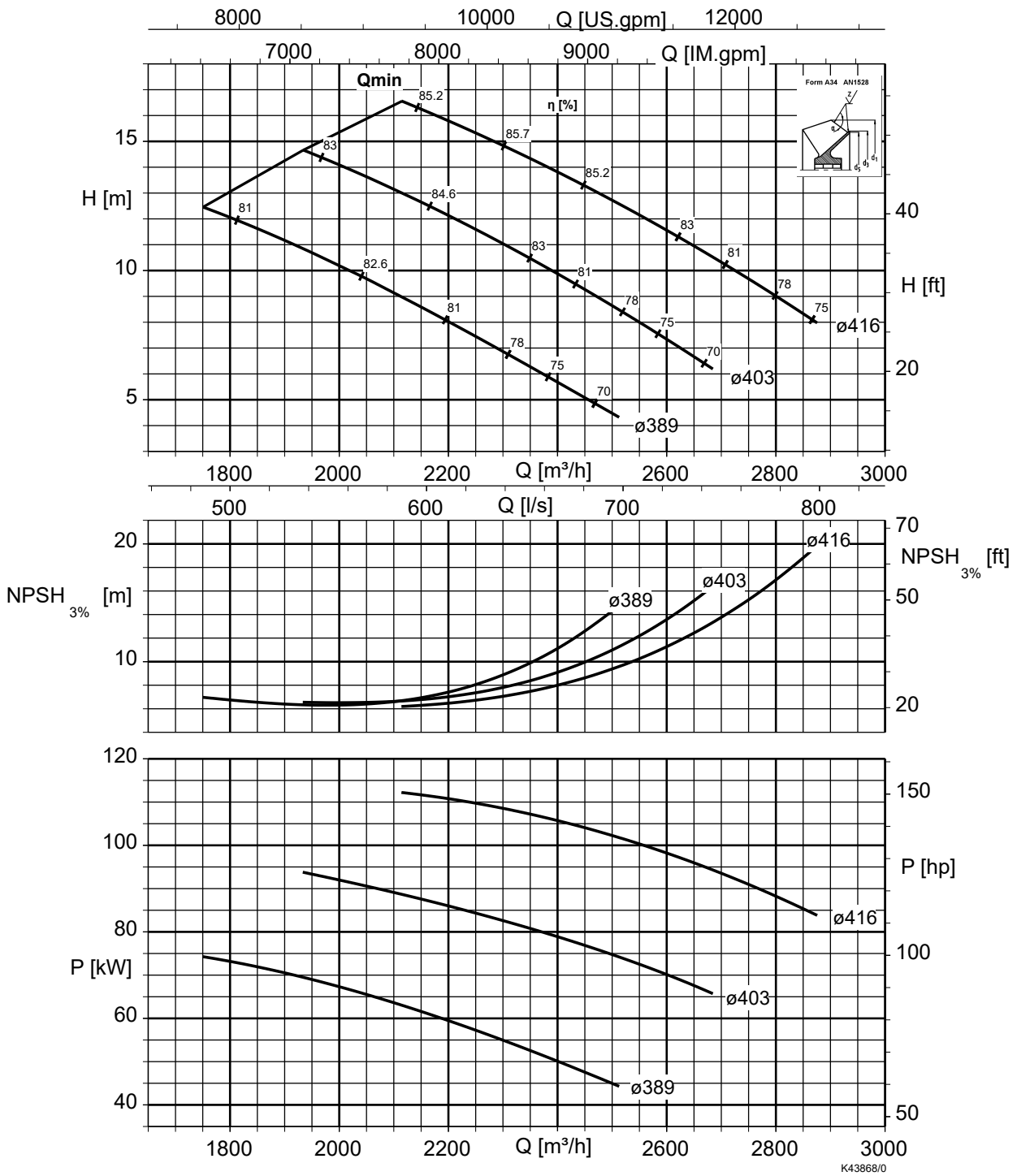
Table 14: Motor data

Size	Motor	P2 [kW]		J
		₋₂₃₎	IE3	[kgm²]
600-420	554 UV/XV	55	-	0,76
600-420	604 UV/XV	60	-	0,78
600-420	704 UT/XT	70	-	0,91
600-420	904 UT/XT	90	-	1,05
600-420	1054UT/XT		55	1,13
600-420	1304UT/XT	-	75	1,21

23 - = No efficiency classification

AmaCan DB3 600-420, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

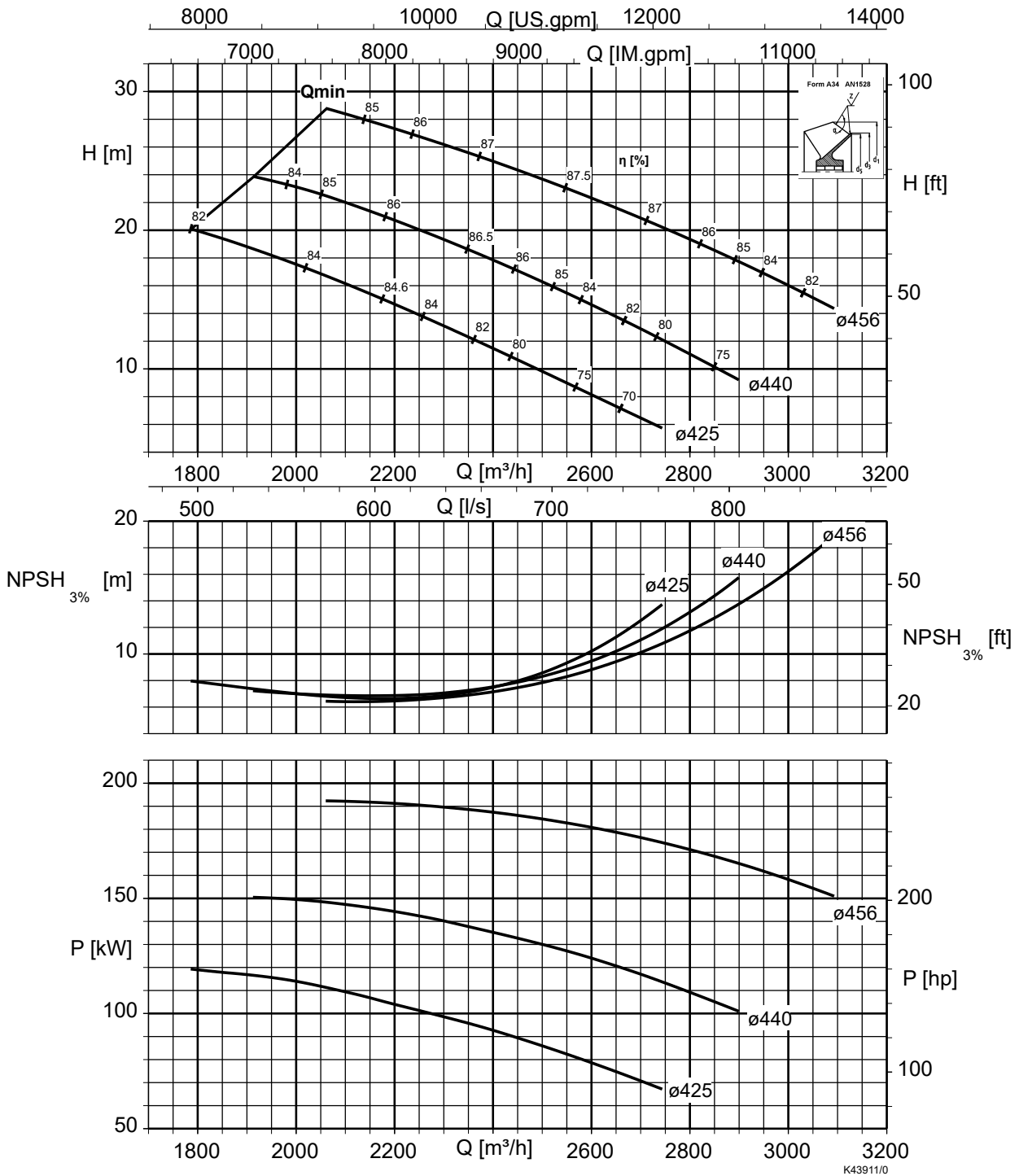
Table 15: Motor data

Size	Motor	P2 [kW]		J
		₋₂₄₎	IE3	[kgm ²]
600-420	904 UT/XT	90	-	1,05
600-420	1054UT/XT	105	-	1,13
600-420	1304UT/XT	130	75	1,21

24 - = No efficiency classification

AmaCan DA3 700/800-460, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

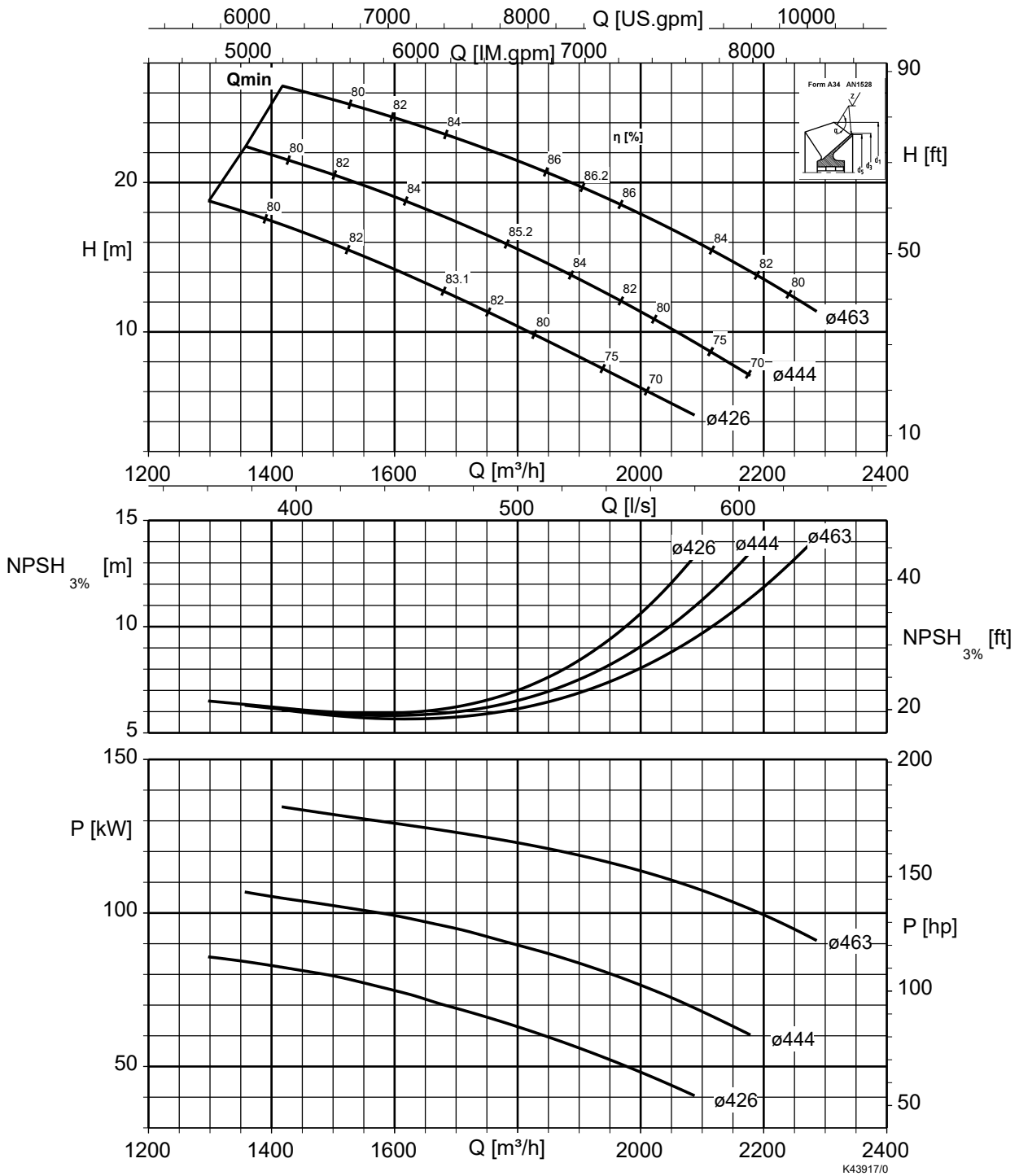
Table 16: Motor data

Size	Motor	P2 [kW]		J
		₋₂₅₎	IE3	[kgm²]
700-460	1304UT/XT	130	-	1,88
700-460	1704UT/XT	170	-	2,52
700-460	1904UT/XT	190	-	2,69
700-460	2154UT/XT	215	-	2,83
800-460	2754UT/XT	-	150	4,86
800-460	3004UT/XT	-	215	5,21

25 - = No efficiency classification

AmaCan DB2 700/800-460, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

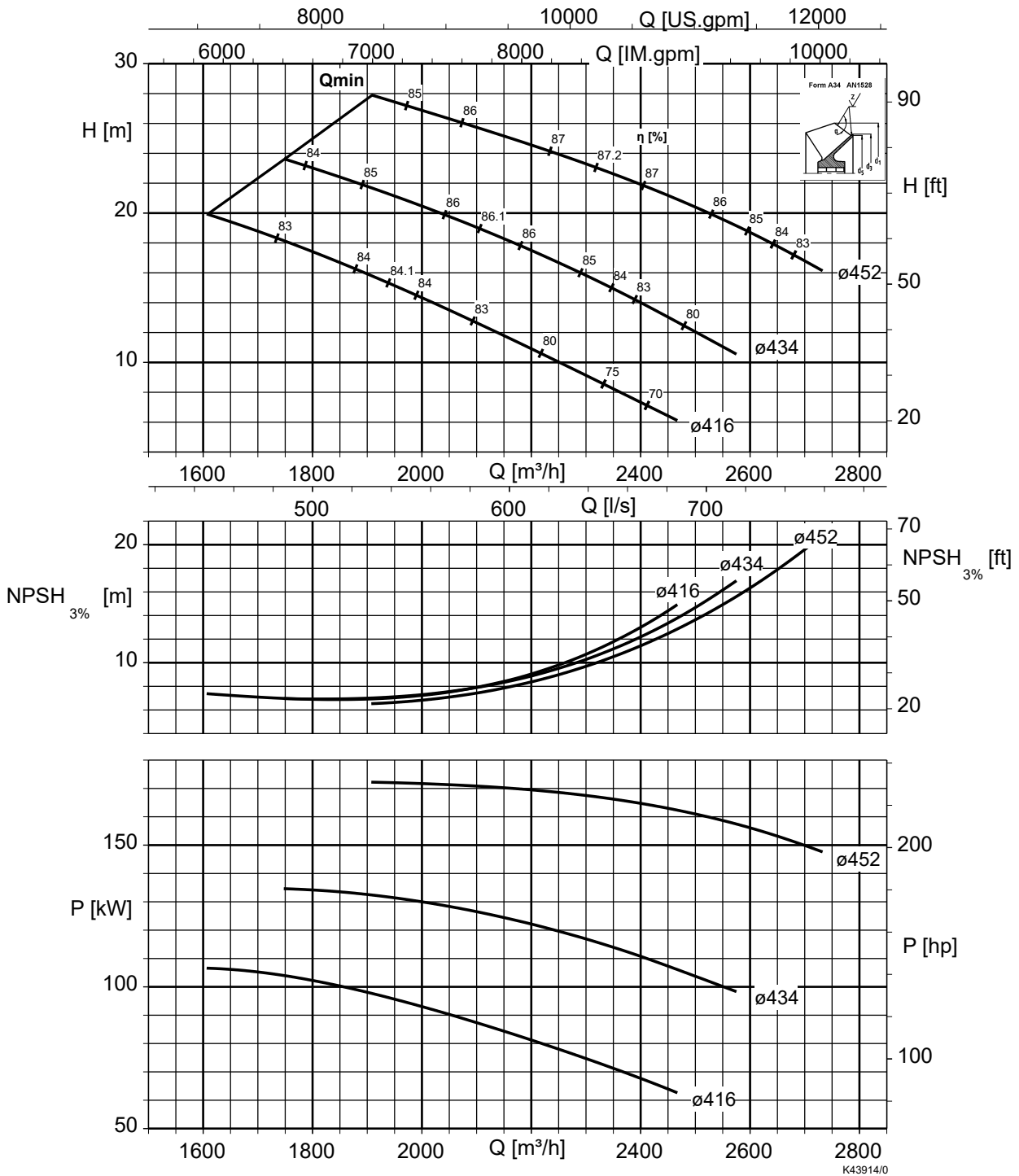
Table 17: Motor data

Size	Motor	P2 [kW]		J
		₂₆₎	IE3	[kgm²]
700-460	904UT/XT	90	-	1,55
700-460	1054UT/XT	105	-	1,63
700-460	1304UT/XT	130	-	1,71
700-460	1504UT/XT	150	-	2,34
700-460	1704UT/XT	170	90	2,34
700-460	1904UT/XT	-	110	2,51
800-460	2754UT/XT	-	150	4,68
800-460	3004UT/XT	-	215	5,03

26 - = No efficiency classification

AmaCan DB3 700/800-460, n = 1450 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

Table 18: Motor data

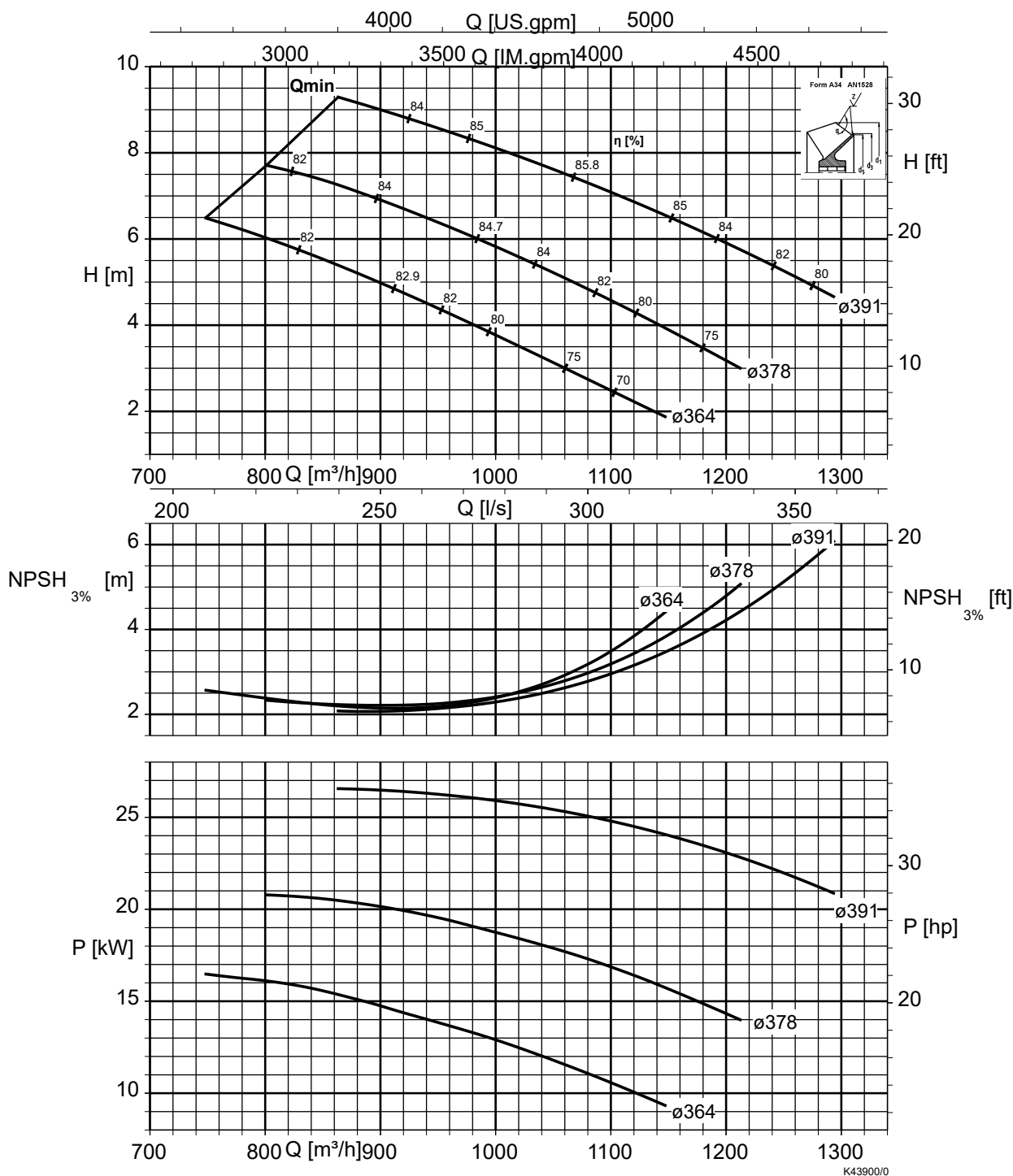
Size	Motor	P2 [kW]		J [kgm²]
		„27)	IE3	
700-460	1304UT/XT	130	-	1,78
700-460	1504UT/XT	150	-	2,41
700-460	1704UT/XT	170	-	2,41
700-460	1904UT/XT	190	-	2,58
700-460	2154UT/XT	215	-	2,72
800-460	2754UT/XT	-	150	4,75
800-460	3004UT/XT	-	215	5,1

27 - = No efficiency classification

$n = 960 \text{ rpm}$

AmaCan DA3 600-390, $n = 960 \text{ rpm}$

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 65 mm

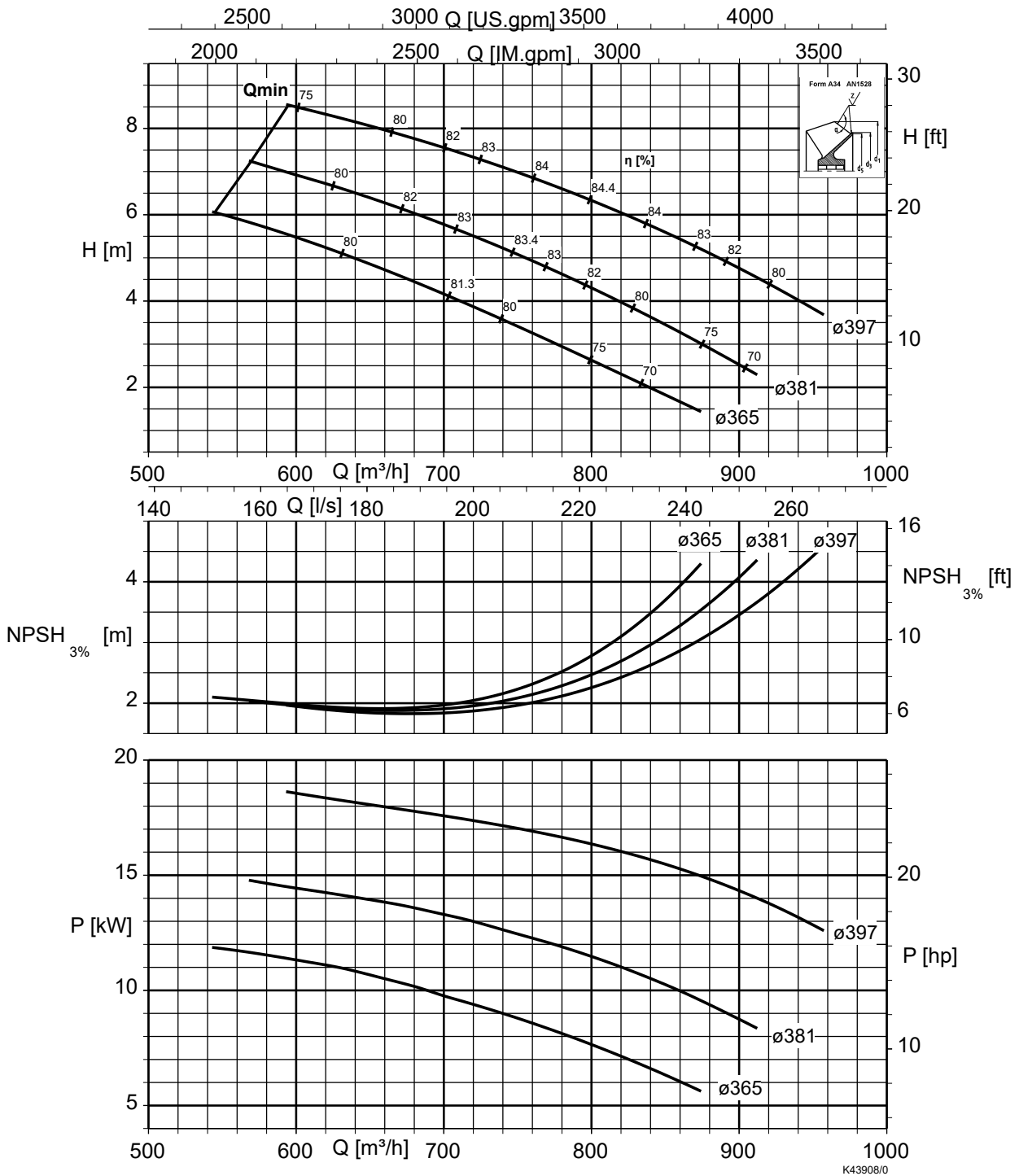
Table 19: Motor data

Size	Motor	P2 [kW]		J
		₋₂₈₎	IE3	[kgm²]
600-390	226UV/XV	22	-	0,77
600-390	306UV/XV	30	-	0,77
600-390	476UT/XT	-	18,5	1,06
600-390	656UT/XT	-	20	1,14
600-390	806UT/XT	-	30	1,25

28 - = No efficiency classification

AmaCan DB2 600-390, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 65 mm

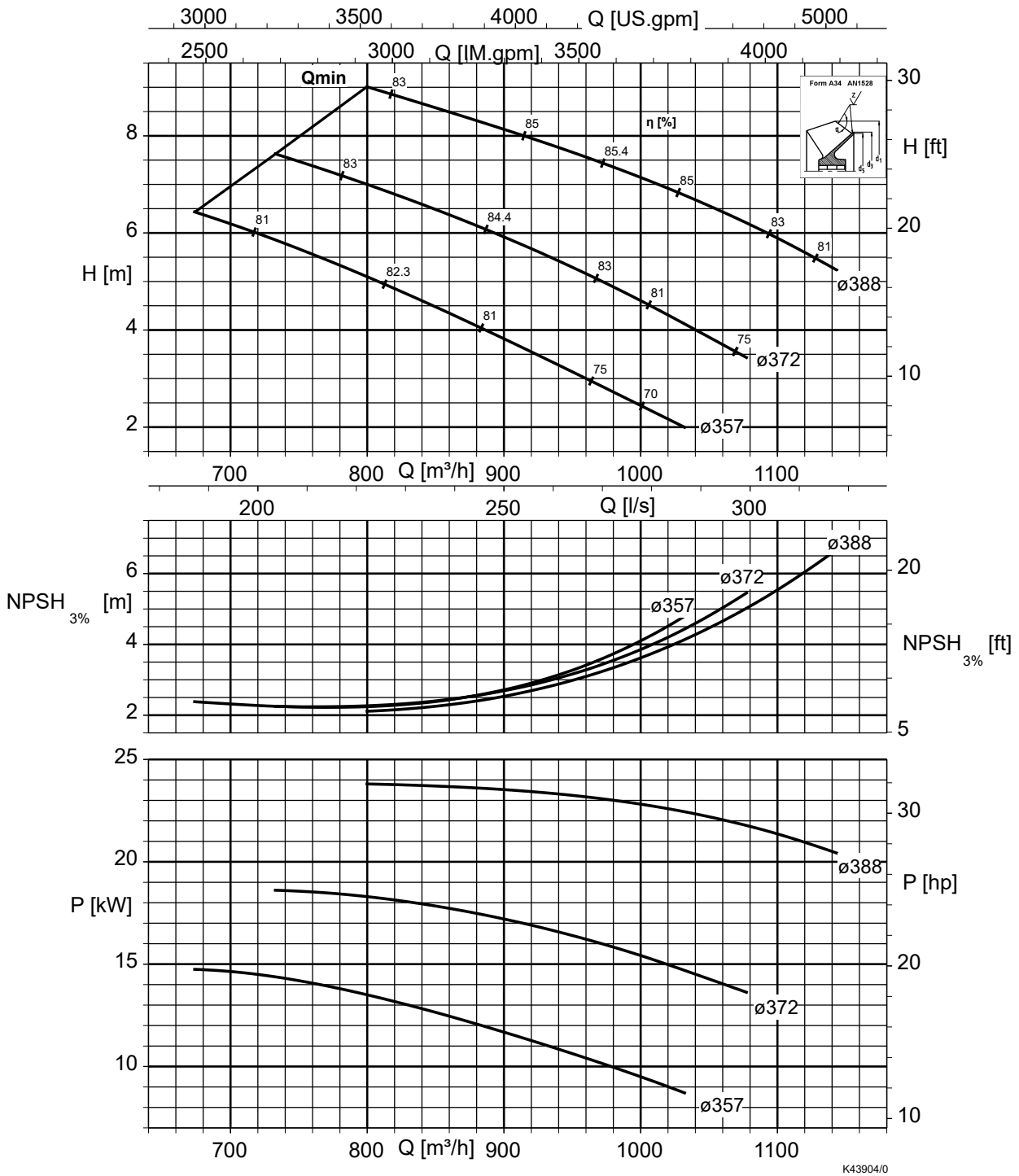
Table 20: Motor data

Size	Motor	P2 [kW]		J
		²⁹⁾	IE3	[kgm ²]
600-390	226UV/XV	22	-	0,68
600-390	476UT/XT	-	18,5	0,98
600-390	656UT/XT	-	22	1,06

29 - = No efficiency classification

AmaCan DB3 600-390, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 65 mm

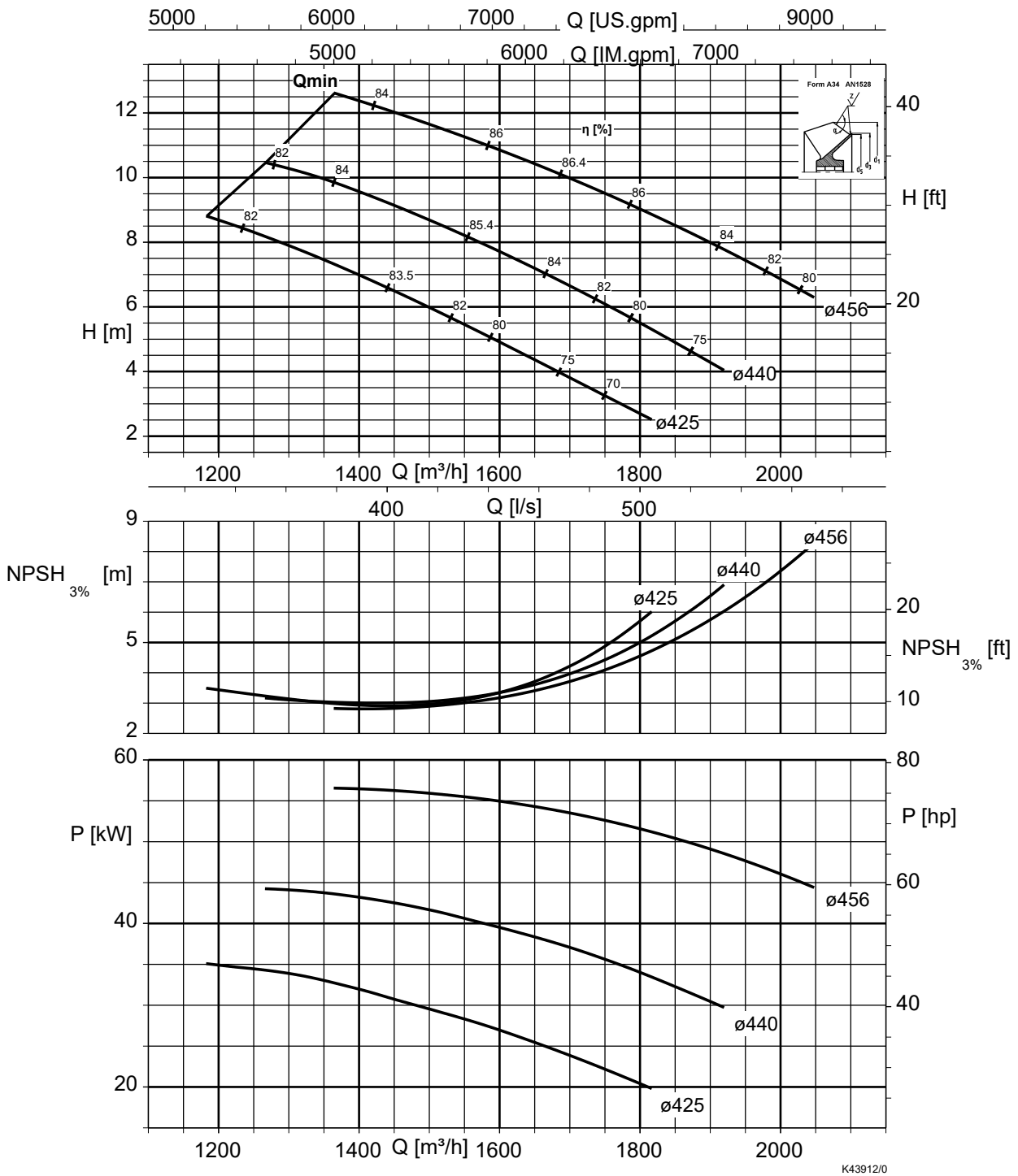
Table 21: Motor data

Size	Motor	P2 [kW]		J
		₋₃₀₎	IE3	[kgm ²]
600-390	226UV/XV	22	-	0,72
600-390	306UV/XV	30	-	0,72
600-390	476UT/XT	-	18,5	1,02
600-390	656UT/XT	-	22	1,1
600-390	806UT/XT	-	30	1,20

30 - = No efficiency classification

AmaCan DA3 700-460, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

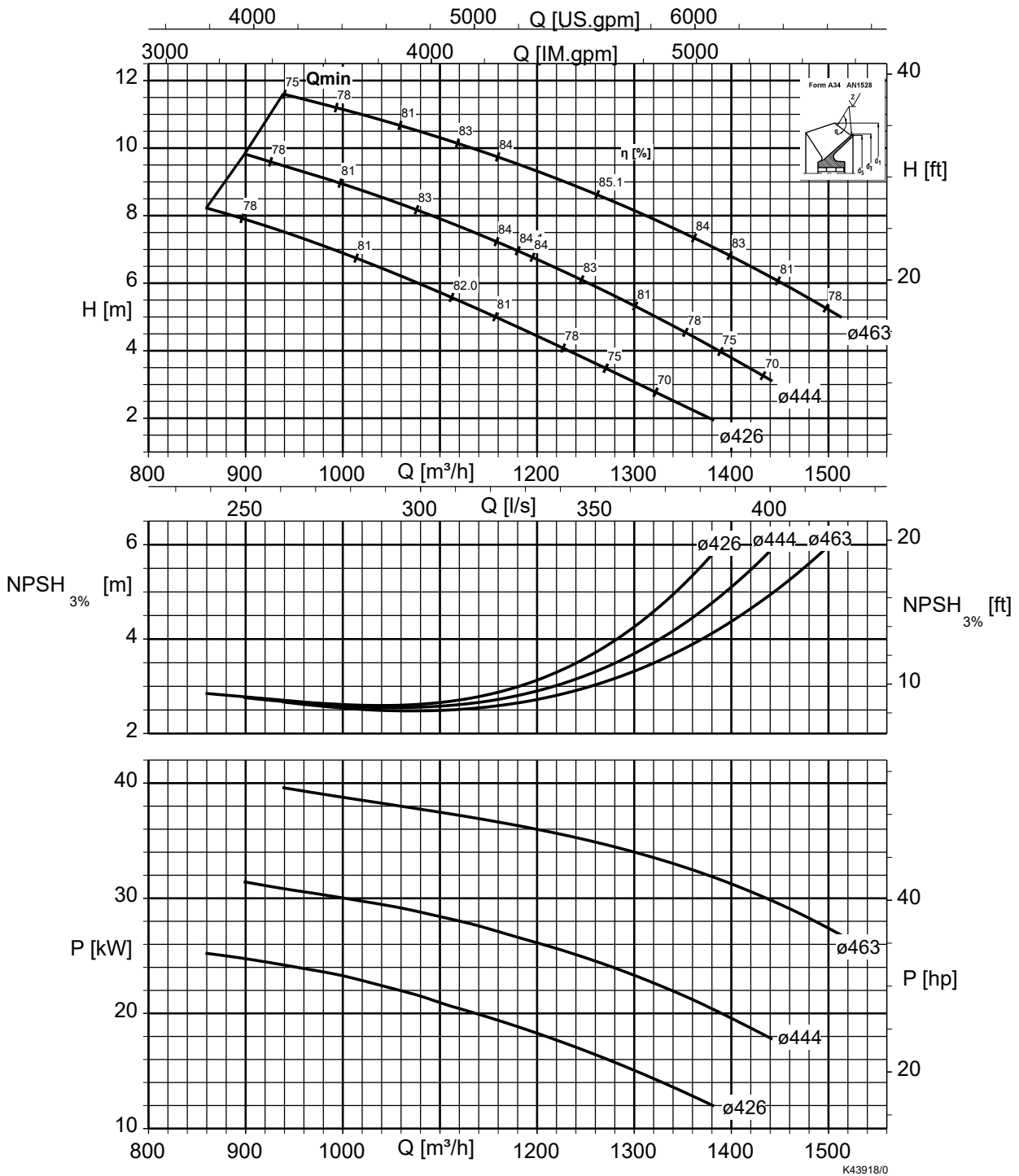
Table 22: Motor data

Size	Motor	P2 [kW]		J
		³¹⁾	IE3	[kgm²]
700-460	476UT/XT	47	-	1,73
700-460	606UT/XT	60	-	1,73
700-460	806UT/XT	80	-	1,92
700-460	1006UT/XT	-	37	2,05
700-460	1206UT/XT	-	45	2,19

31 - = No efficiency classification

AmaCan DB2 700-460, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

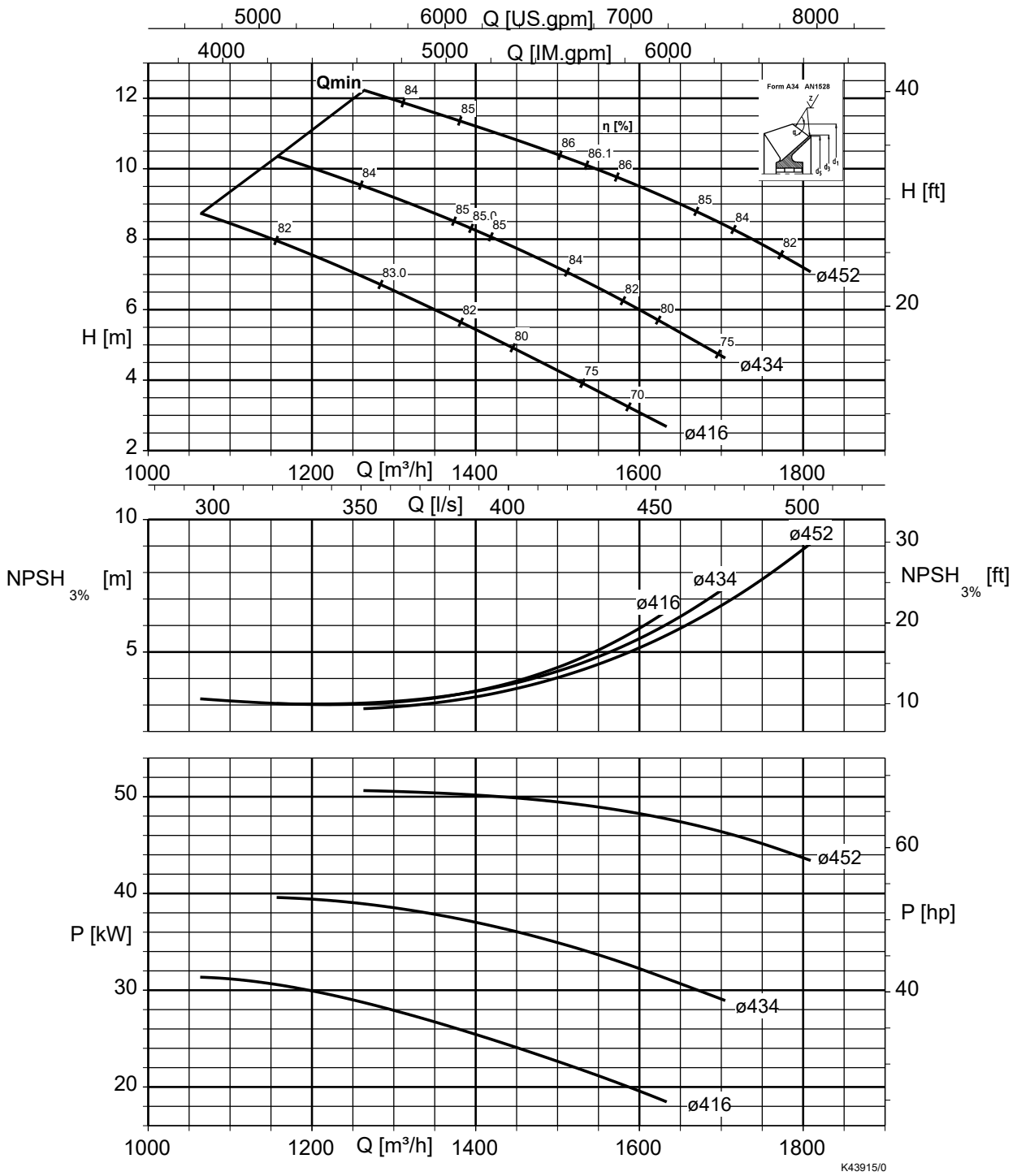
Table 23: Motor data

Size	Motor	P2 [kW]		J
		³²⁾	IE3	[kgm²]
700-460	476UT/XT	47	-	1,56
700-460	806UT/XT	-	30	1,75
700-460	1006UT/XT	-	37	1,88
700-460	1206UT/XT	-	45	2,02

32 - = No efficiency classification

AmaCan DB3 700-460, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 76 mm

Table 24: Motor data

Size	Motor	P2 [kW]		J
		³³⁾	IE3	[kgm ²]
700-460	476UT/XT	47	-	1,63
700-460	606UT/XT	60	-	1,63
700-460	1006UT/XT	-	37	1,95
700-460	1206UT/XT	-	45	2,09

33 - = No efficiency classification

AmaCan DA3 800-580, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed

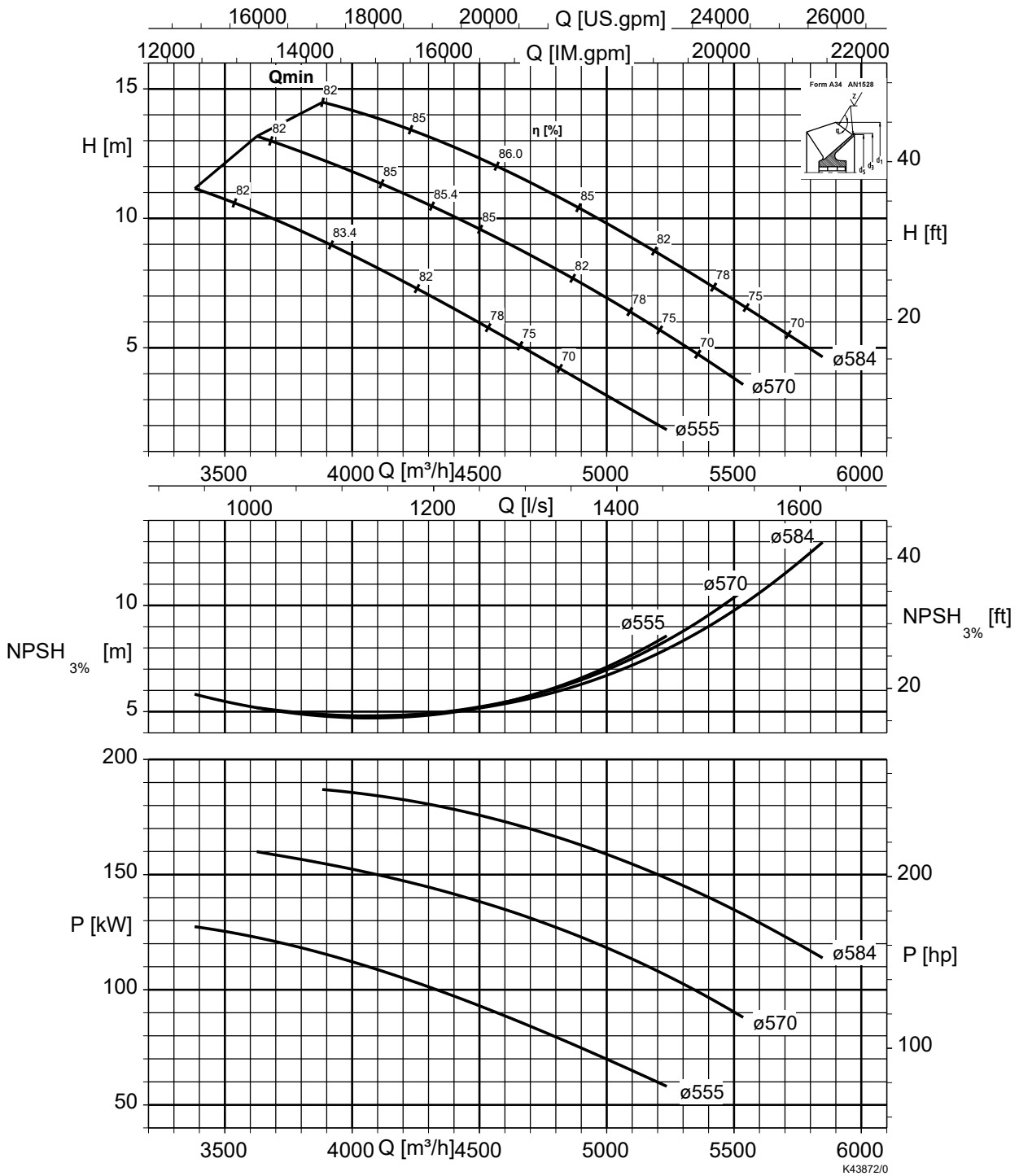


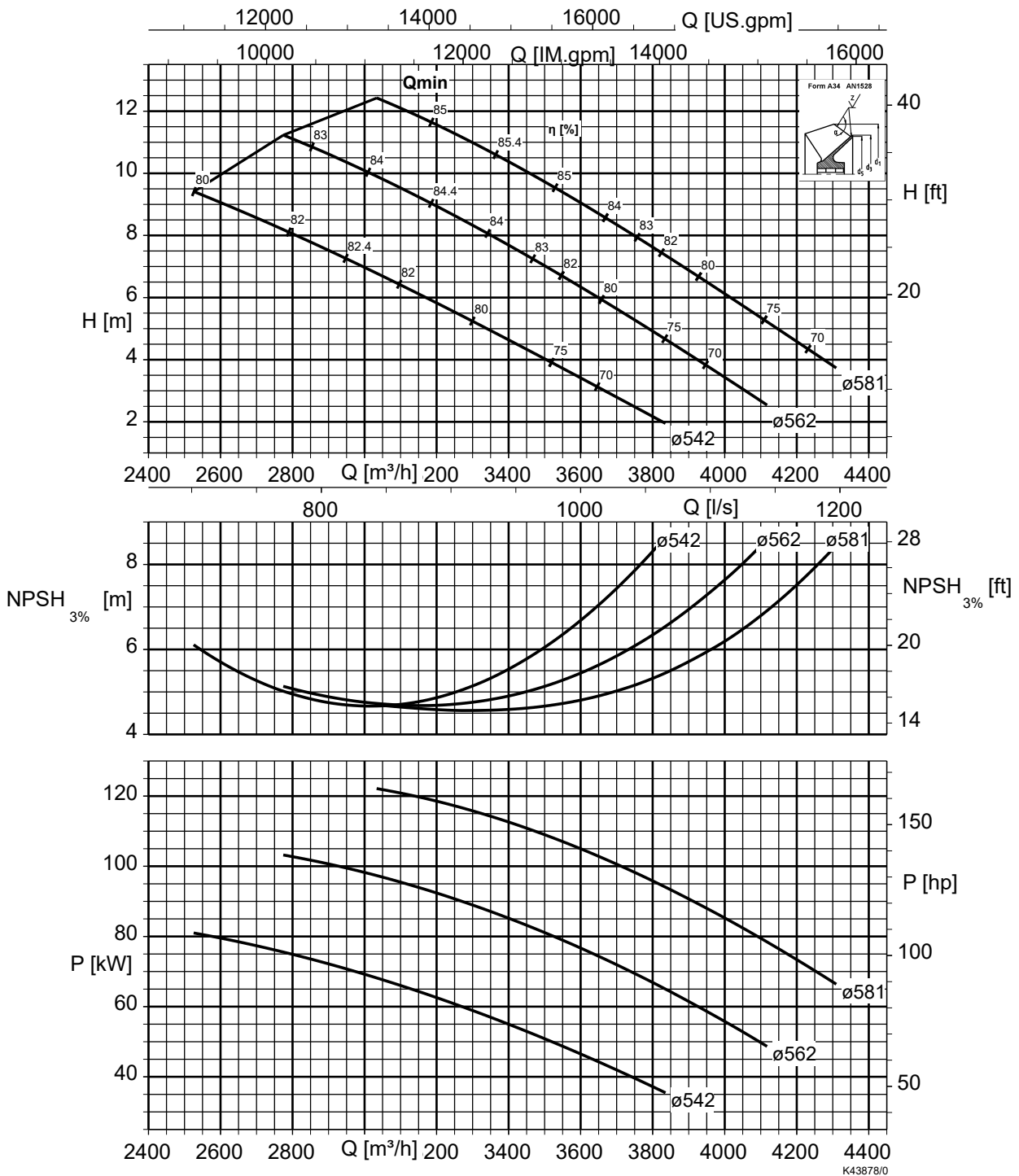
Table 25: Motor data

Size	Motor	P2 [kW]		J
		³⁴⁾	IE3	[kgm²]
800-580	1556UT/XT	155	-	5,61
800-580	1806UT/XT	180	-	5,89
800-580	2056UT/XT	205	-	6,18
800-580	2506UT/XT	-	135	10,75
800-580	2906UT/XT	-	150	12,02
800-580	3406UT/XT	-	180	13,29

34 - = No efficiency classification

AmaCan DB2 800-580, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 105 mm

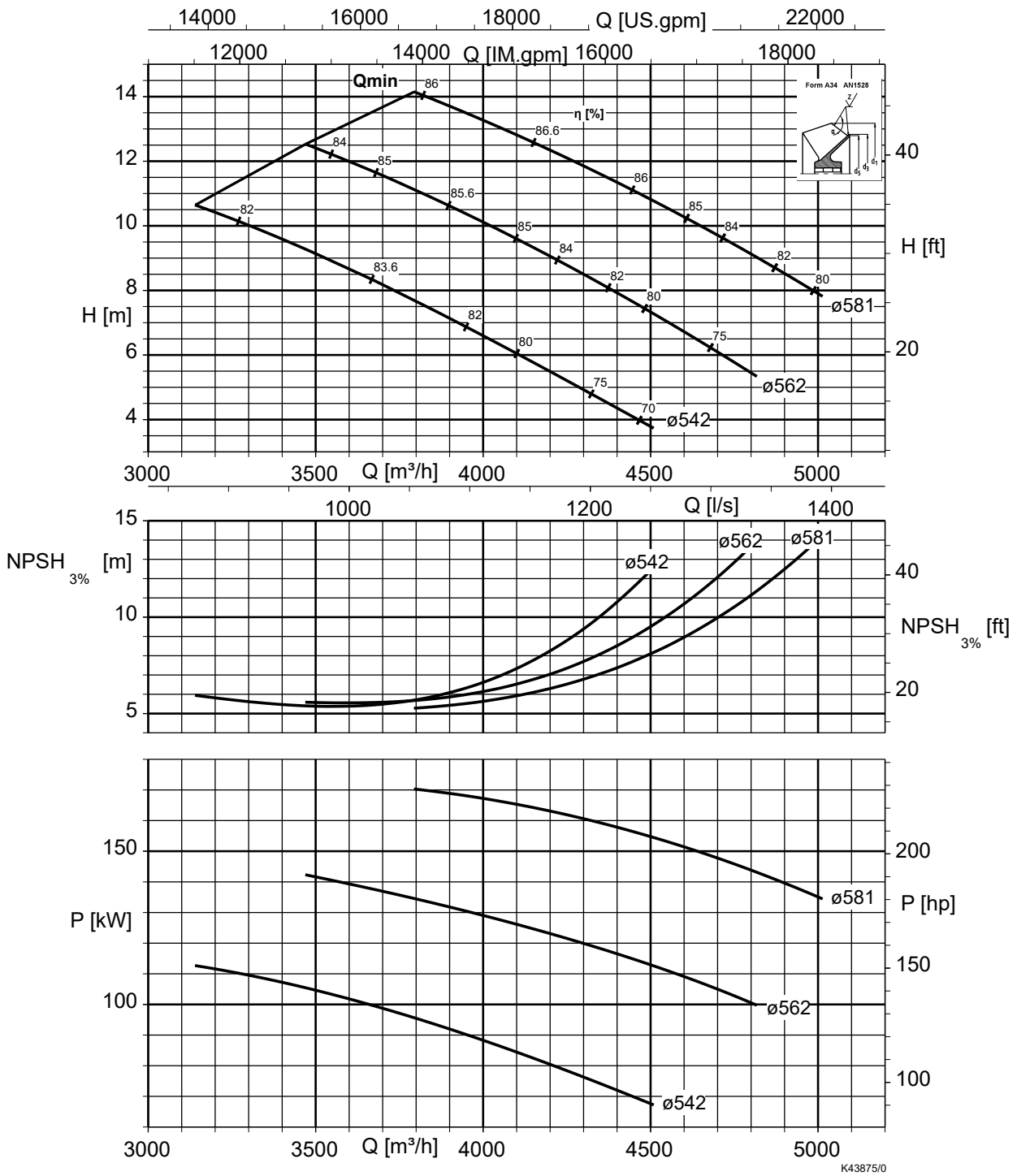
Table 26: Motor data

Size	Motor	P2 [kW]		J
		³⁵⁾	IE3	[kgm²]
800-580	806UT/XT	80	-	3,86
800-580	1006UT/XT	100	-	3,99
800-580	1206UT/XT	115	-	4,13
800-580	1556UT/XT	155	-	5,17
800-580	2506UT/XT	-	135	10,31
800-580	2906UT/XT	-	150	11,58

35 - = No efficiency classification

AmaCan DB3 800-580, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 105 mm

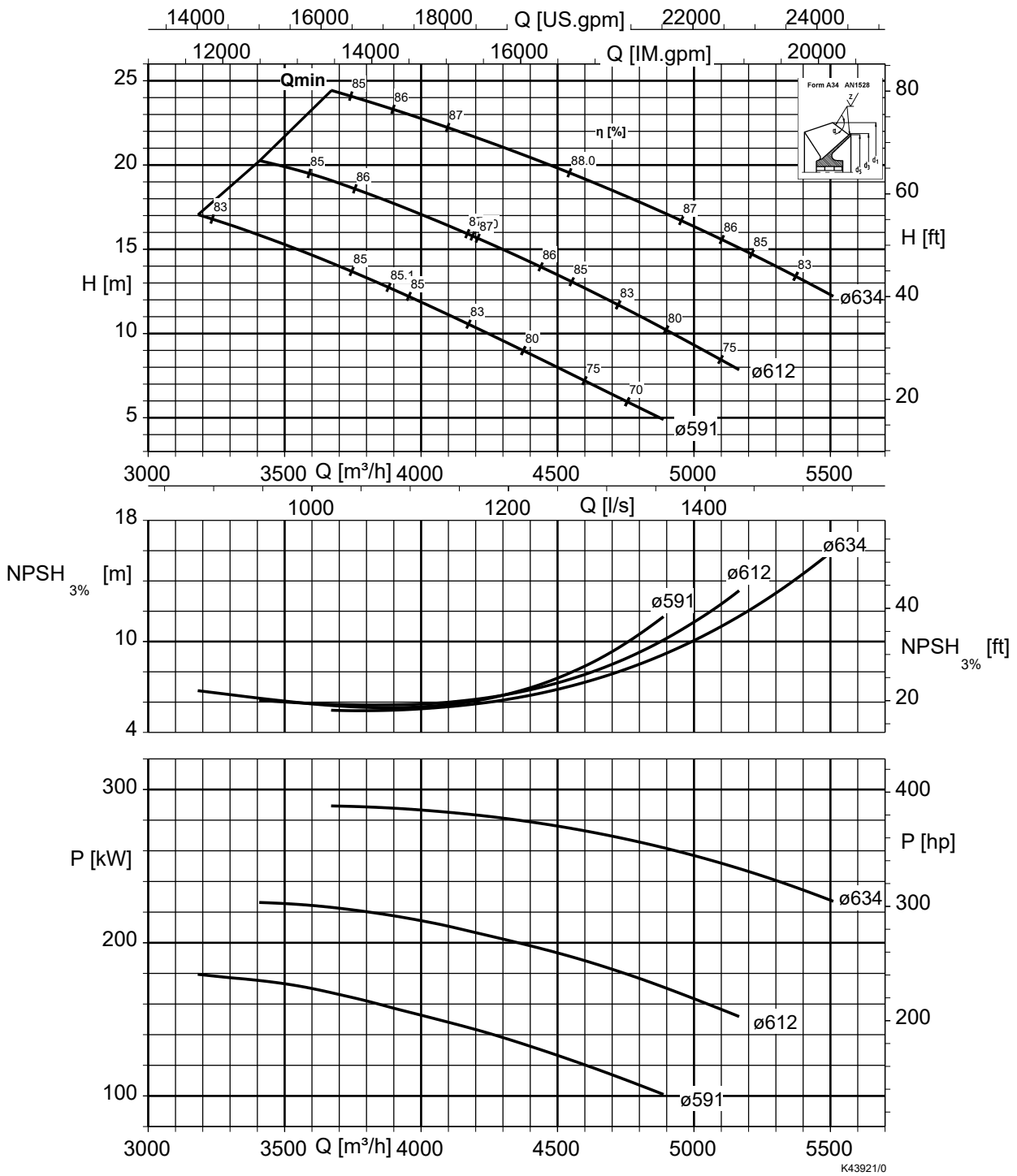
Table 27: Motor data

Size	Motor	P2 [kW]		J
		³⁶⁾	IE3	[kgm²]
800-580	1206UT/XT	115	-	4,12
800-580	1556UT/XT	155	-	5,17
800-580	1806UT/XT	180	-	5,45
800-580	2056UT/XT	205	-	5,74
800-580	2506UT/XT	-	135	10,31
800-580	2906UT/XT	-	150	11,58
800-580	3406UT/XT	-	180	12,85

36 - = No efficiency classification

AmaCan DA3 900-630, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 105 mm

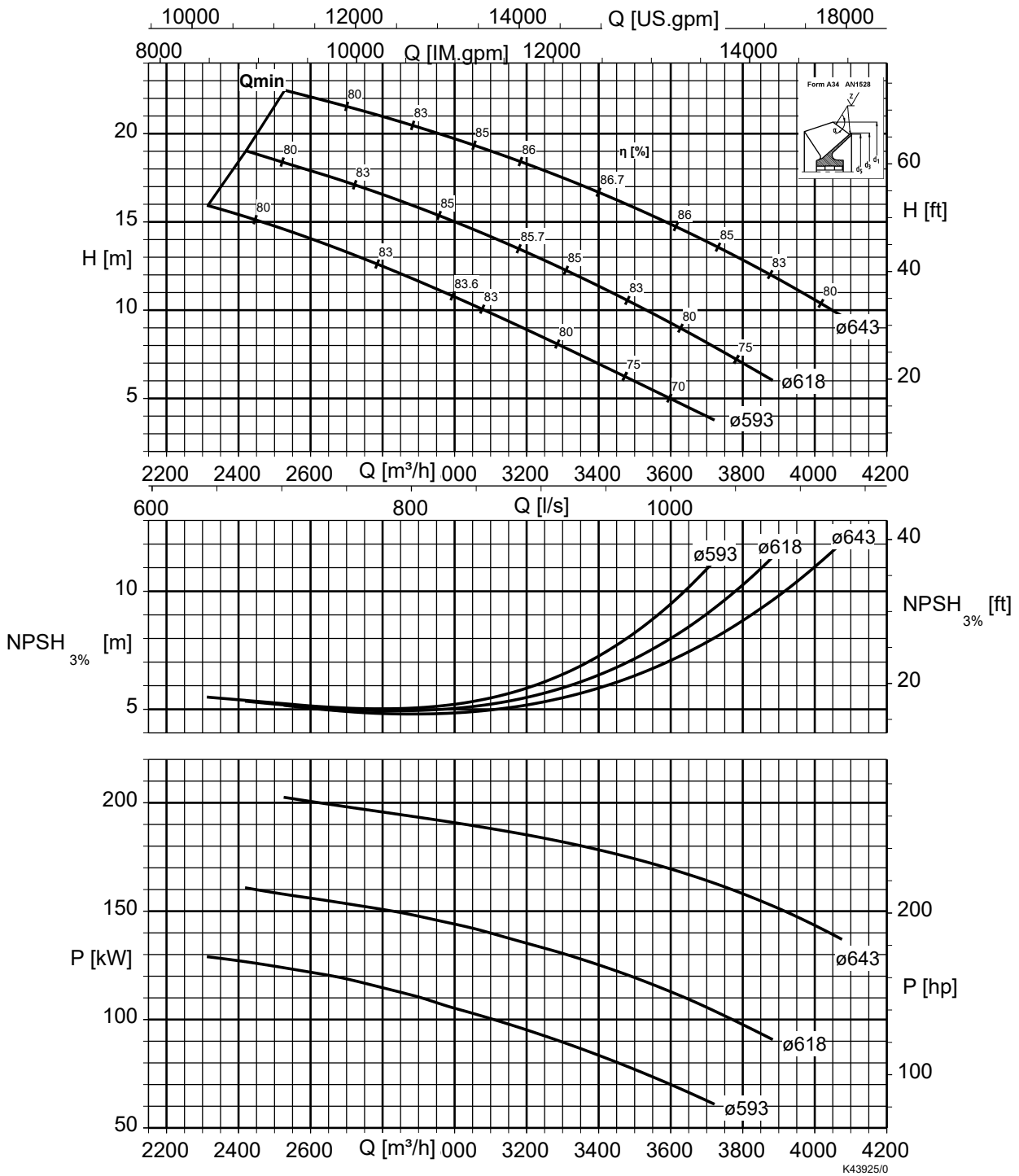
Table 28: Motor data

Size	Motor	P2 [kW]		J
		³⁷⁾	IE3	[kgm²]
900-630	1806UT/XT	180	-	9,77
900-630	2056UT/XT	205	-	10,07
900-630	2506UT/XT	250	-	14,87
900-630	2906UT/XT	290	-	16,14
900-630	3406UT/XT	340	180	17,41

37 - = No efficiency classification

AmaCan DB2 900-630, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 105 mm

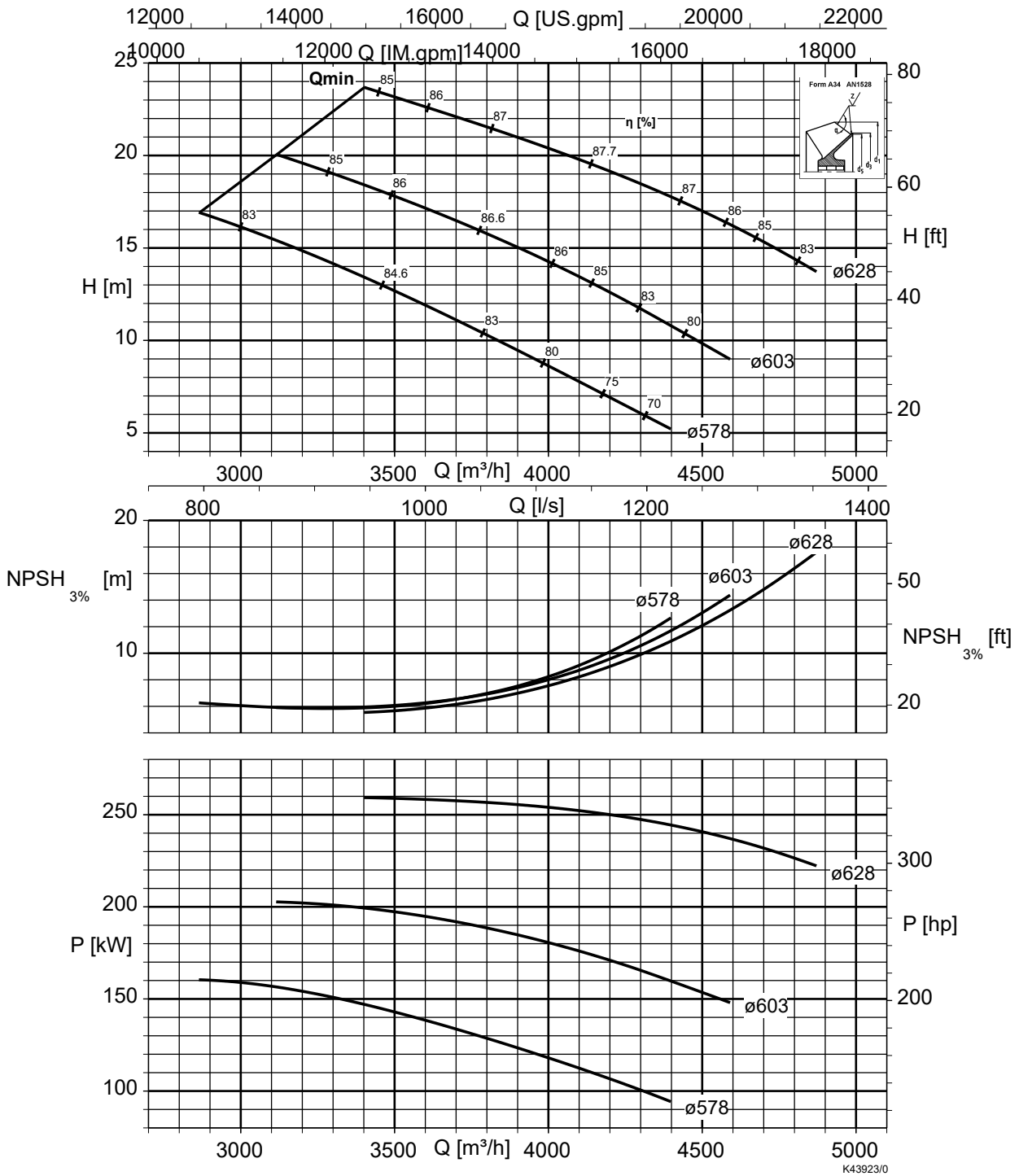
Table 29: Motor data

Size	Motor	P2 [kW]		J
		³⁸⁾	IE3	[kgm²]
900-630	1556UT/XT	155	-	7,34
900-630	1806UT/XT	180	-	7,61
900-630	2056UT/XT	205	-	7,91
900-630	2506UT/XT	250	135	12,71
900-630	2906UT/XT	-	150	13,98
900-630	3406UT/XT	-	180	15,25

38 - = No efficiency classification

AmaCan DB3 900-630, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 105 mm

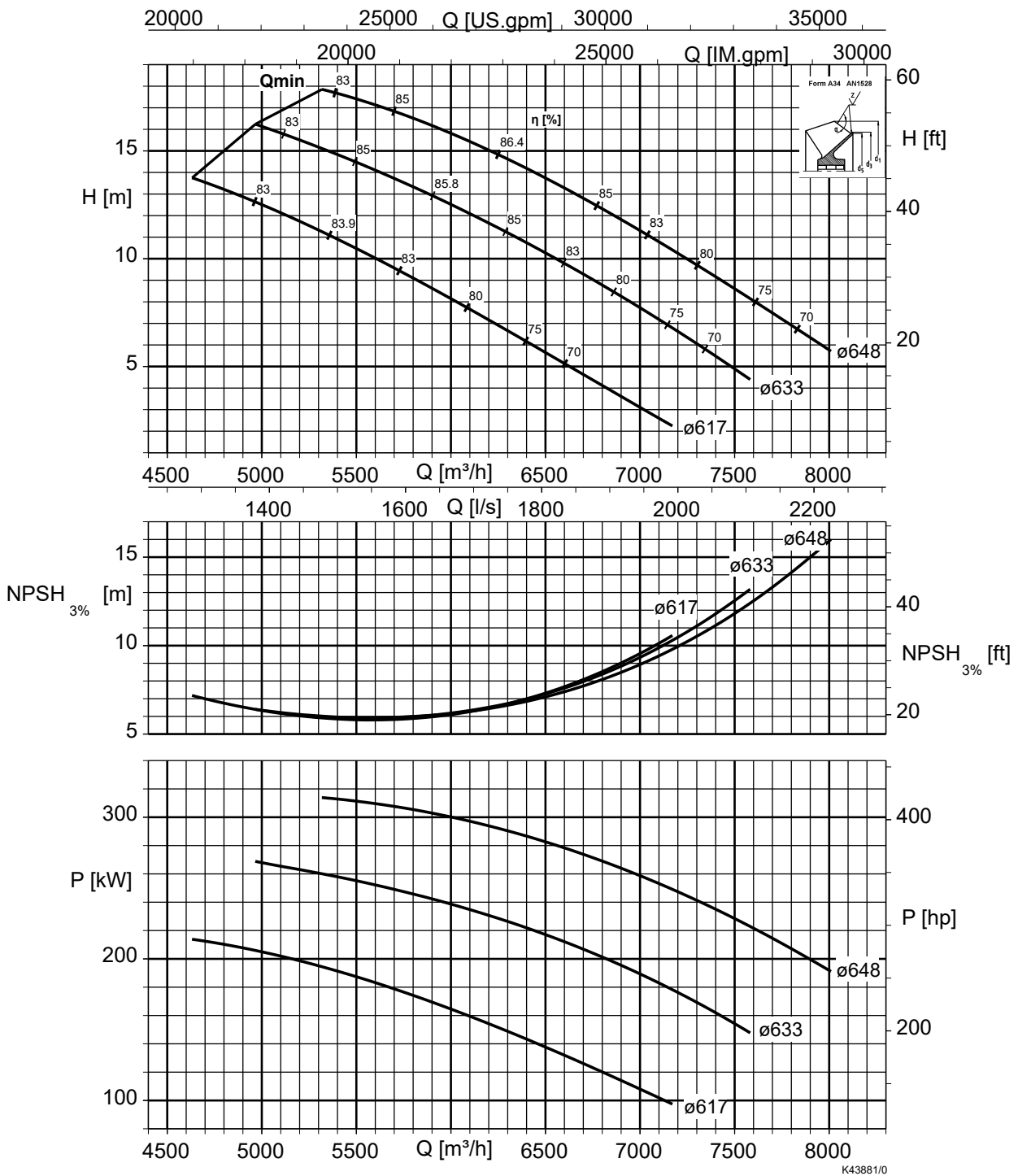
Table 30: Motor data

Size	Motor	P2 [kW]		J
		³⁹⁾	IE3	[kgm²]
900-630	1806UT/XT	180	-	7,8
900-630	2056UT/XT	205	-	8,1
900-630	2506UT/XT	250	-	12,9
900-630	2906UT/XT	290	-	14,17
900-630	3406UT/XT	340	180	15,44

39 - = No efficiency classification

AmaCan DA3 900-650, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 116 mm

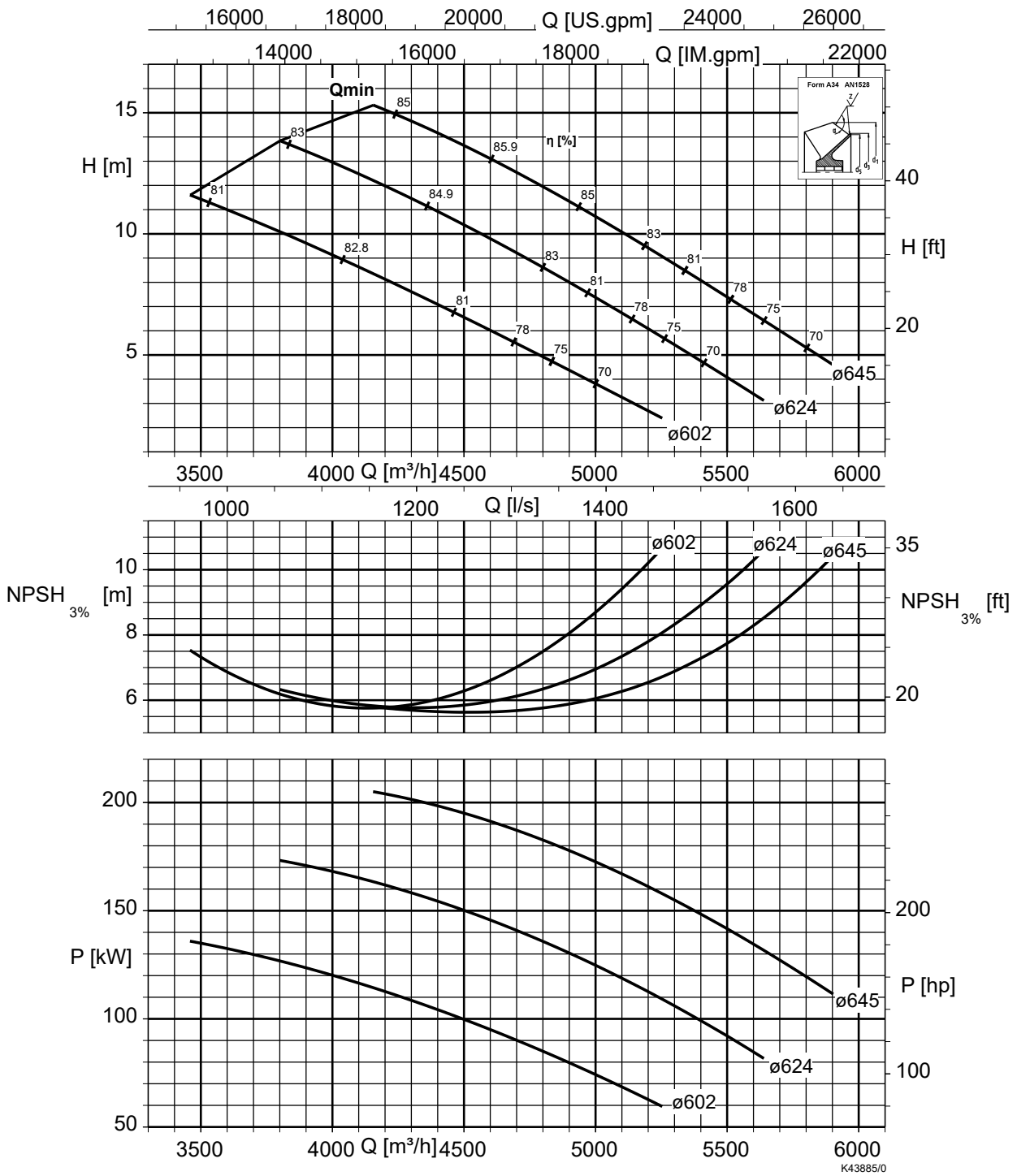
Table 31: Motor data

Size	Motor	P2 [kW]		J
		⁴⁰⁾	IE3	[kgm ²]
900-650	2506UT/XT	250	-	13,51
900-650	2906UT/XT	290	-	14,78
900-650	3406UT/XT	340	-	16,05

40 - = No efficiency classification

AmaCan DB2 900-650, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 116 mm

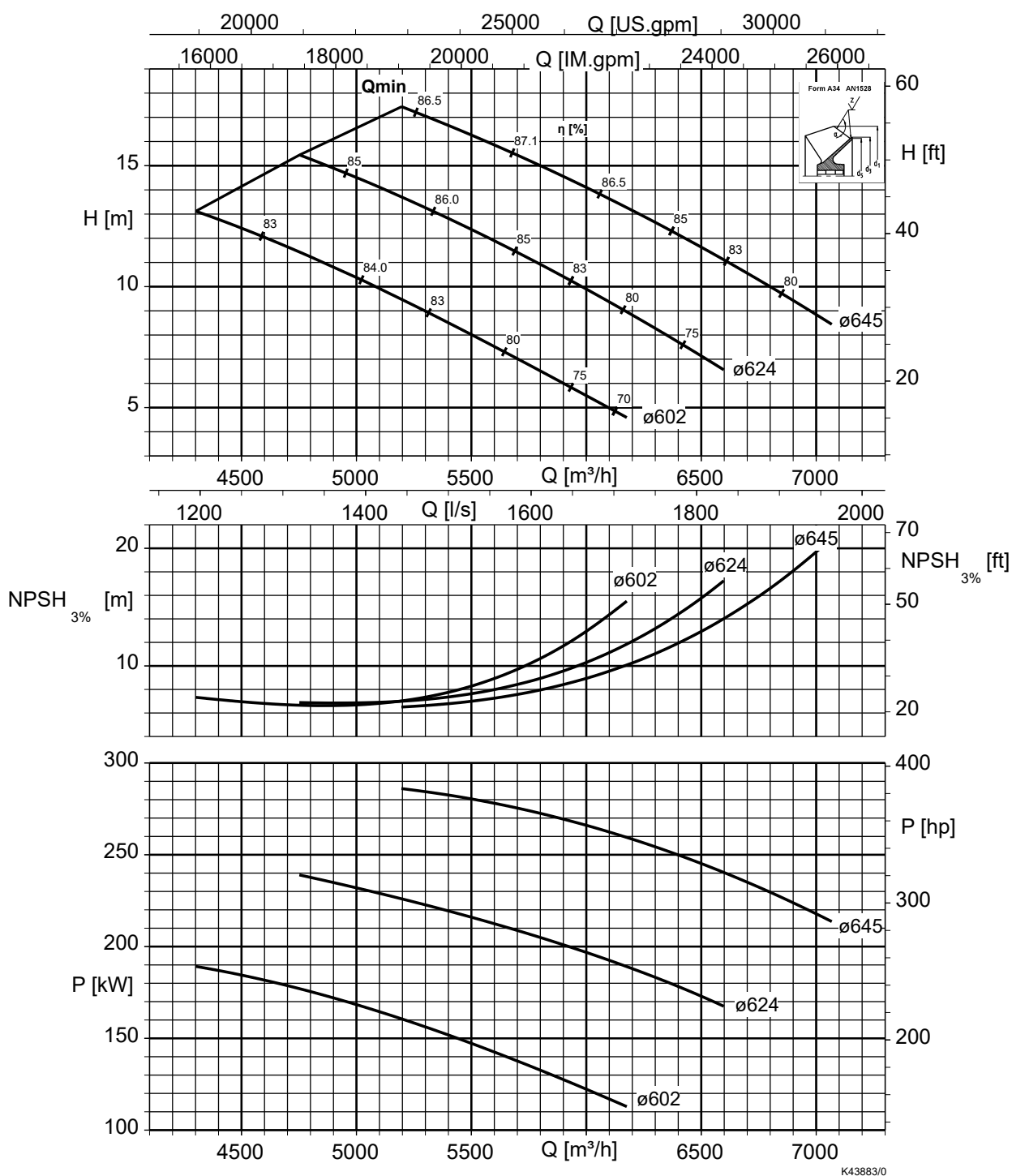
Table 32: Motor data

Size	Motor	P2 [kW]		J
		⁴¹⁾	IE3	[kgm²]
900-650	1556UT/XT	155	-	7,38
900-650	1806UT/XT	180	-	7,65
900-650	2056UT/XT	205	-	7,95
900-650	2506UT/XT	250	135	12,75
900-650	2906UT/XT	-	150	14,02
900-650	3406UT/XT	-	180	15,29

41 - = No efficiency classification

AmaCan DB3 900-650, n = 960 rpm

Characteristic curves in acc. with ISO 9906 / 2 / 2B. n = speed



Free passage = 116 mm

Table 33: Motor data

Size	Motor	P2 [kW]		J
		⁴²⁾	IE3	[kgm²]
900-650	2056UT/XT	205	-	7,95
900-650	2506UT/XT	250	-	12,75
900-650	2906UT/XT	290	-	14,02
900-650	3406UT/XT	340	-	15,29

42 - = No efficiency classification

Dimensions [mm]

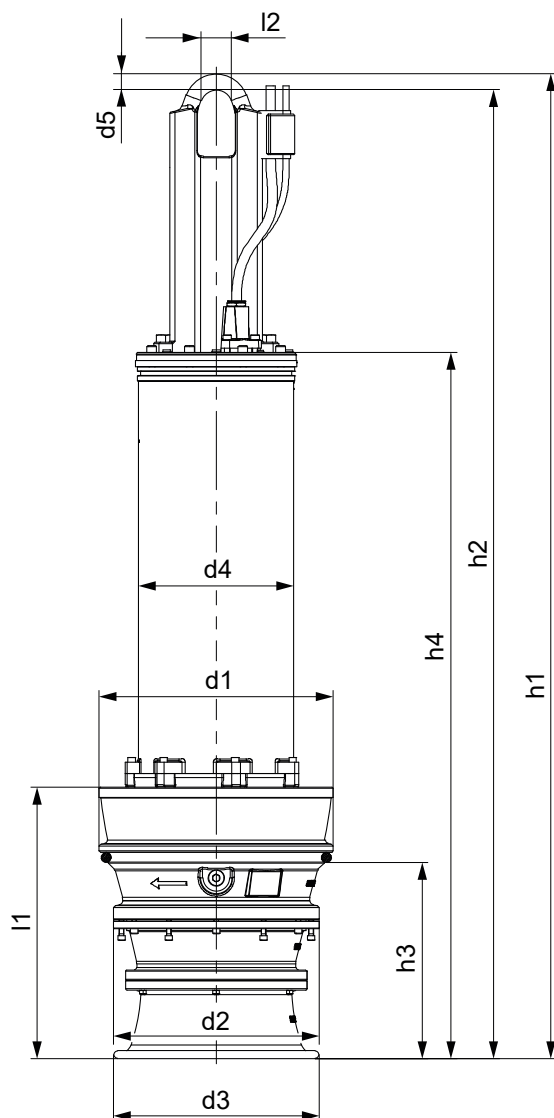


Fig. 1: Pump set dimensions

Table 34: Pump set dimensions [mm]

Size	Motor size	Number of poles	d ₁	d ₂	d ₃	d ₄	d ₅	h ₁	h ₂	h ₃	h ₄	l ₁	l ₂	[kg] ⁴³⁾
600-390	35	4	582	500	530	303	23	1939	1916	517	1511	722	70	575
600-390	45	4	582	500	530	303	23	1939	1916	517	1511	722	70	575
600-390	55	4	582	500	530	303	23	1939	1916	517	1511	722	70	575
600-390	60	4	582	500	530	303	23	1939	1916	517	1511	722	70	575
600-390	70	4	582	500	530	385	40	2279	2239	517	1588	722	80	945
600-390	80	4	582	500	530	385	40	2279	2239	517	1588	722	80	970
600-390	90	4	582	500	530	385	40	2479	2439	517	1788	722	80	1070
600-390	105	4	582	500	530	385	40	2479	2439	517	1788	722	80	1110
600-390	130	4	582	500	530	385	40	2479	2439	517	1788	722	80	1150
600-390	22	6	582	500	530	303	23	1939	1916	517	1511	722	70	575
600-390	30	6	582	500	530	303	23	1939	1916	517	1511	722	70	575
600-390	47	6	582	500	530	385	40	2279	2239	517	1588	722	80	945
600-390	65	6	582	500	530	385	40	2279	2239	517	1588	722	80	960
600-390	80	6	582	500	530	385	40	2479	2439	517	1788	722	80	1060

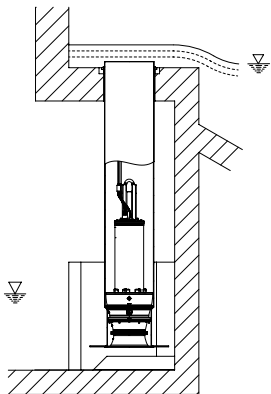
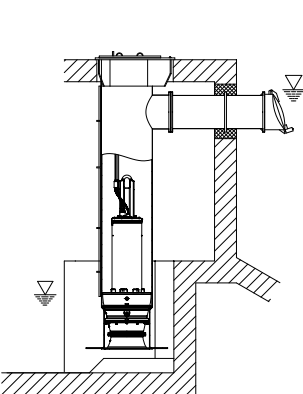
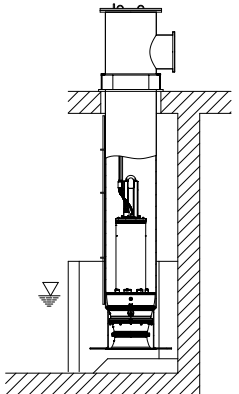
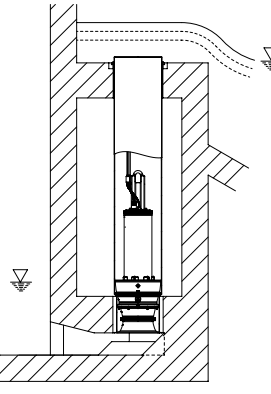
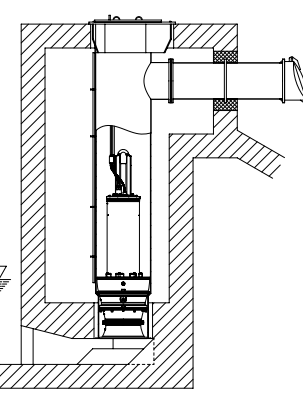
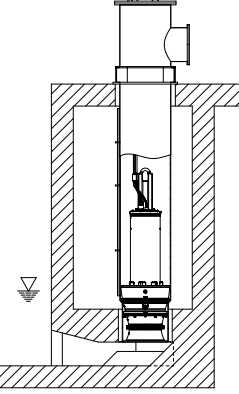
⁴³ Pump set in material variant G3, with 10-metre power cable and 5-metre support rope The indicated weights are reference values only. Refer to the data sheet for the exact weight.

Size	Motor size	Num- ber of poles	d ₁	d ₂	d ₃	d ₄	d ₅	h ₁	h ₂	h ₃	h ₄	l ₁	l ₂	[kg] ⁽⁴³⁾
700-390	170	4	582	500	530	475	40	2800	2760	517	2109	722	80	1555
700-390	190	4	582	500	530	475	40	2800	2760	517	2109	722	80	1605
600-420	55	4	580	510	510	303	23	1903	1880	486	1475	673	70	605
600-420	60	4	580	510	510	303	23	1903	1880	486	1475	673	70	620
600-420	70	4	580	510	510	385	40	2243	2203	486	1552	673	80	905
600-420	90	4	580	510	510	385	40	2443	2403	486	1752	673	80	1045
600-420	105	4	580	510	510	385	40	2443	2403	486	1752	673	80	1085
600-420	130	4	580	510	510	385	40	2443	2403	486	1752	673	80	1125
700-420	170	4	580	510	510	475	40	2764	2724	486	2073	673	80	1530
700-420	190	4	580	510	510	475	40	2764	2724	486	2073	673	80	1580
700-420	215	4	580	510	510	475	40	2764	2724	486	2073	673	80	1610
700-460	90	4	678	580	610	385	40	2550	2510	553	1859	807	80	1130
700-460	105	4	678	580	610	385	40	2550	2510	553	1859	807	80	1175
700-460	130	4	678	580	610	385	40	2550	2510	553	1859	807	80	1215
700-460	150	4	678	580	610	475	40	2724	2684	553	2033	807	80	1625
700-460	170	4	678	580	610	475	40	2724	2684	553	2033	807	80	1625
700-460	190	4	678	580	610	475	40	2724	2684	553	2033	807	80	1670
700-460	215	4	678	580	610	475	40	2724	2684	553	2033	807	80	1700
700-460	47	6	678	580	610	385	40	2350	2310	553	1659	807	80	1020
700-460	60	6	678	580	610	385	40	2350	2310	553	1659	807	80	1020
700-460	80	6	678	580	610	385	40	2550	2510	553	1859	807	80	1145
700-460	100	6	678	580	610	385	40	2550	2510	553	1859	807	80	1190
700-460	120	6	678	580	610	385	40	2550	2510	553	1859	807	80	1235
700-460	155	6	678	580	610	475	40	2724	2684	553	2033	807	80	1605
800-460	275	4	678	580	610	555	50	3130	3071	553	2531	807	90	2310
800-460	300	4	678	580	610	555	50	3130	3071	553	2531	807	90	2375
800-580	80	6	778	695	705	385	40	2603	2563	636	1912	900	80	1335
800-580	100	6	778	695	705	385	40	2603	2563	636	1912	900	80	1380
800-580	120	6	778	695	705	385	40	2603	2563	636	1912	900	80	1425
800-580	155	6	778	695	705	475	40	2777	2737	636	2086	900	80	1795
800-580	180	6	778	695	705	475	40	2777	2737	636	2086	900	80	1825
800-580	205	6	778	695	705	475	40	2777	2737	636	2086	900	80	1880
800-580	250	6	778	695	705	555	50	3184	3124	636	2584	900	90	2530
800-580	290	6	778	695	705	555	50	3184	3124	636	2584	900	90	2670
800-580	340	6	778	695	705	555	50	3184	3124	636	2584	900	90	2800
900-630	155	6	876	790	790	475	40	2928	2888	593	2237	1085	80	2110
900-630	180	6	876	790	790	475	40	2928	2888	593	2237	1085	80	2140
900-630	205	6	876	790	790	475	40	2928	2888	593	2237	1085	80	2195
900-630	250	6	876	790	790	555	50	3188	3129	593	2589	1085	90	2835
900-630	290	6	876	790	790	555	50	3188	3129	593	2589	1085	90	2975
900-630	340	6	876	790	790	555	50	3188	3129	593	2589	1085	90	3105
900-650	155	6	876	760	770	475	40	2872	2832	680	2181	1028	80	2015
900-650	180	6	876	760	770	475	40	2872	2832	680	2181	1028	80	2040
900-650	205	6	876	760	770	475	40	2872	2832	680	2181	1028	80	2100
900-650	250	6	876	760	770	555	50	3131	3072	680	2532	1028	90	2740
900-650	290	6	876	760	770	555	50	3131	3072	680	2532	1028	90	2875
900-650	340	6	876	760	770	555	50	3131	3072	680	2532	1028	90	3010

Installation types

Nine design variants are available ⁴⁴⁾:

Table 35: Overview of installation types

		
BU ⁴⁵⁾ /BUS ⁴⁶⁾ discharge tube	CU ⁴⁵⁾ /CUS ⁴⁶⁾ discharge tube	DU ⁴⁵⁾ /DUS ⁴⁶⁾ discharge tube
Design with overflow discharge for installation in open intake chamber	Design with underfloor discharge for installation in open intake chamber	Design with above-floor discharge nozzle for installation in open intake chamber
		
BG discharge tube ⁴⁵⁾	CG discharge tube ⁴⁵⁾	DG discharge tube ⁴⁵⁾
Design with overflow discharge for installation in covered intake chamber for low suction-side water levels	Design with underfloor discharge for installation in covered intake chamber for low suction-side water levels	Design with above-floor discharge nozzle for installation in covered intake chamber for low suction-side water levels

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Pump set complete with power cables
- O-ring
- Back-up name plate

Optional accessories:

- Support rope
- Accessories for installing the cable guide:

- Fitting
- Turnbuckle
- Support
- Shackle
- Cable clamps
- Cable support sleeves
- Flow-straightening vane to prevent floor vortices
- Discharge tube

44 For information on the various designs (foundation dimensions, intake chamber, etc.) refer to the general arrangement drawings.

45 Design without suction umbrella

46 Design with suction umbrella

Accessories

Dimensions of the flow-straightening vane

Design of the intake chamber wall surfaces (to prevent vortex formation)

The flow-straightening vane is indispensable for the inlet conditions of the pump set. It prevents the development of a submerged vortex (floor vortex) which could cause a drop in performance, for example. In addition, the floor and wall surfaces of the intake chamber should be designed as a rough concrete surface. Rough surfaces minimise the separation of boundary layers that may cause wall and floor vortices.

Flow-straightening vane and intake chamber

- The anti-swirl baffles in the bellmouth must be aligned with the flow-straightening vane.
- The bail of the pump is oriented in the same direction as the anti-swirl baffles in the bellmouth.

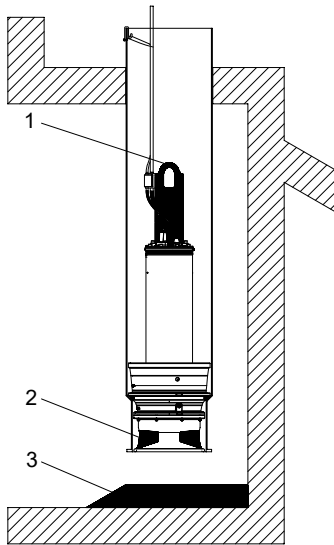
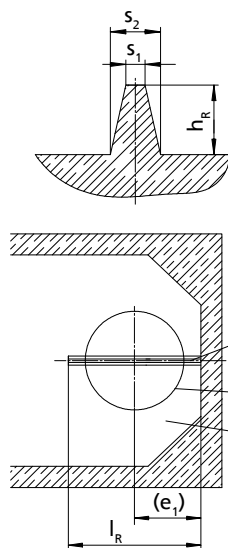


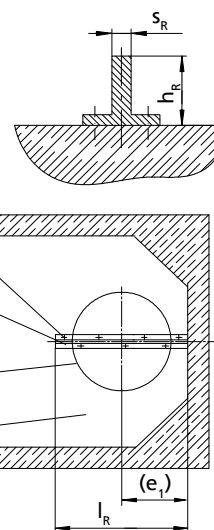
Fig. 2: Installation position of the pump set

1	Bail
2	Anti-swirl baffles
3	Flow-straightening vane

Variant 1
Flow-straightening vane cast from concrete

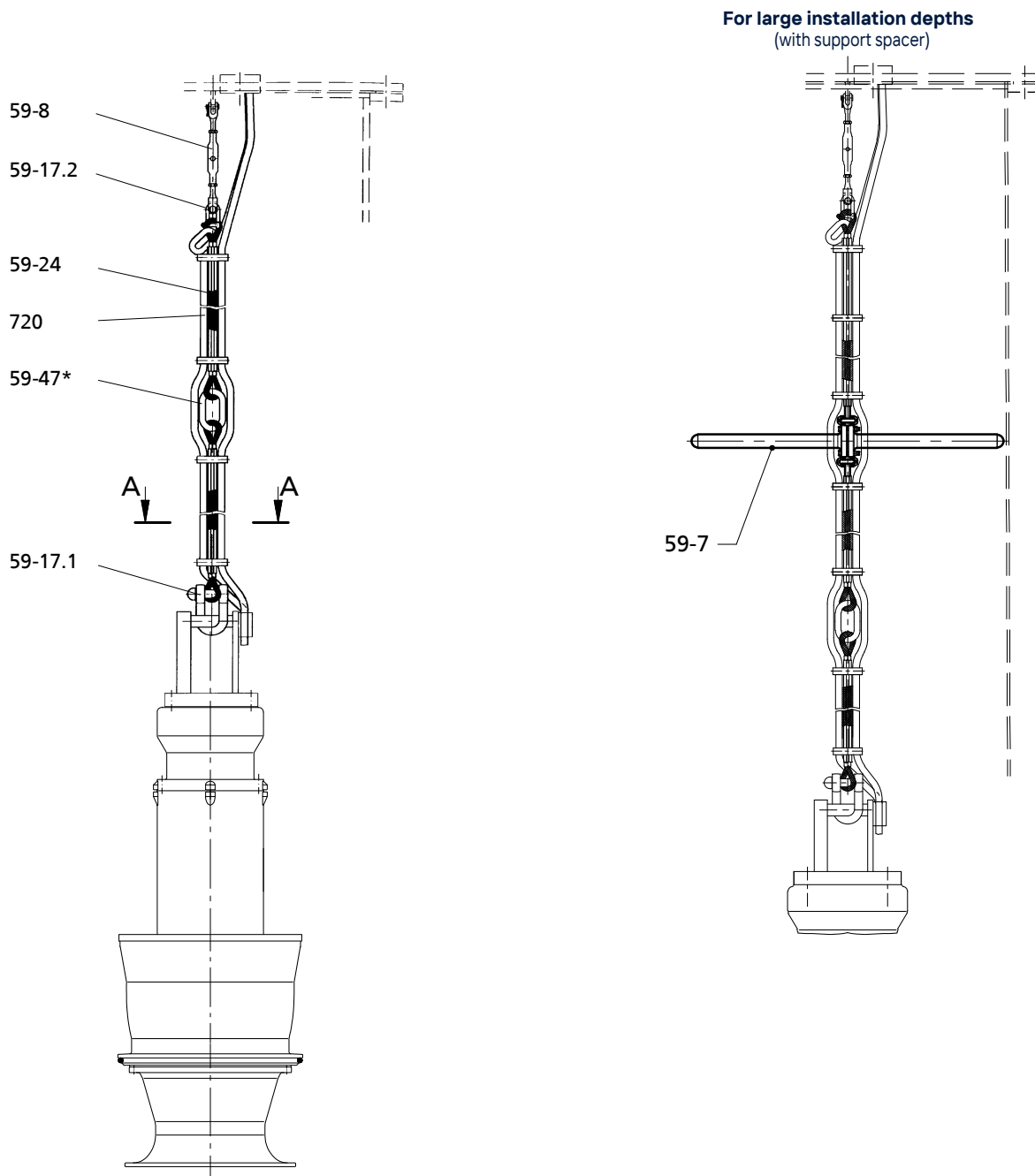


Variant 2
Steel section



A	Bolted to the floor of the intake chamber
B	Flow-straightening vane centred beneath the discharge tube
C	Discharge tube
D	Intake chamber

Support rope and turnbuckle in the discharge tube



*= The number of (intermediate) lifting rings depends on the lifting height of the lifting equipment and on the building structure. (Intermediate lifting rings are supplied as an option).

Table 36: List of components

Part No.	Description	Material
59-8	Turnbuckle	Stainless steel
59-17.2	Shackle	Stainless steel
59-47	(Intermediate) lifting ring(s)	Stainless steel
59-24	Rope, low-rotation design	Stainless steel

Part No.	Description	Material
720	Fitting	EPDM
59-17.1	Shackle	Stainless steel (material option from a total weight of 4250 kg: steel)
59-7	Support	GFRP

Cross-section of cable support

A-A

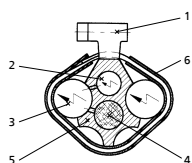


Table 37: List of components

Part No.	Description	Part No.	Description
1	Hose clip	4	Support rope 59-24
2	Control cable	5	Fitting
3	Power cable	6	Clamp cover

Discharge tube cover with cable gland

Design: with welding sleeve

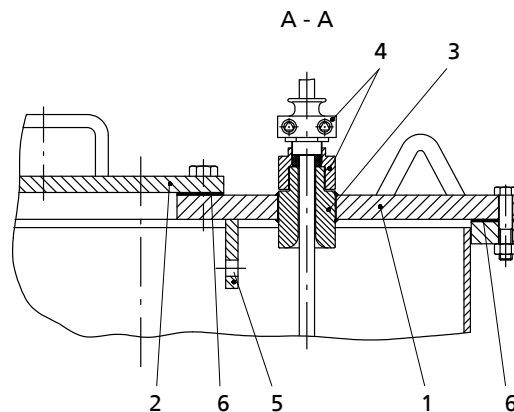
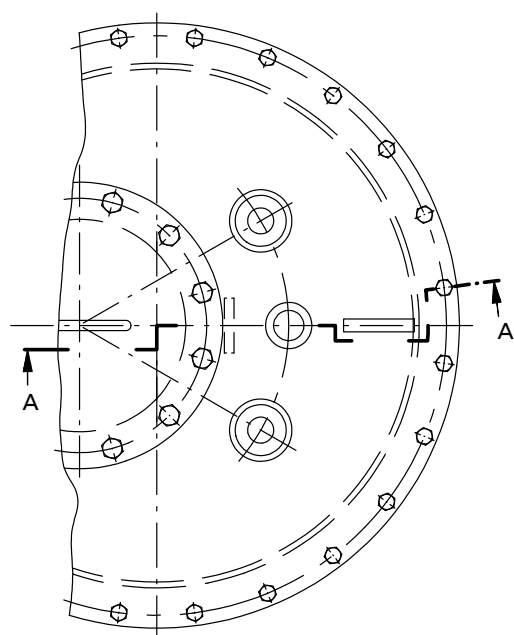


Fig. 3: Design variant with welding sleeve

Table 38: List of components

Part No.	Description	Part No.	Description
1	Discharge tube cover ⁴⁷⁾	4	Threaded bush with cable entry to DIN 22419 with strain relief and protection against kinking and twisting
2	Cover	5	Eyeplate for fastening the cable support (support rope)
3	Welding sleeve	6	Gasket, e.g. fabric-reinforced rubber

⁴⁷ Discharge tube cover also available in split design.

Design variant with transit frame (up to 1 bar)

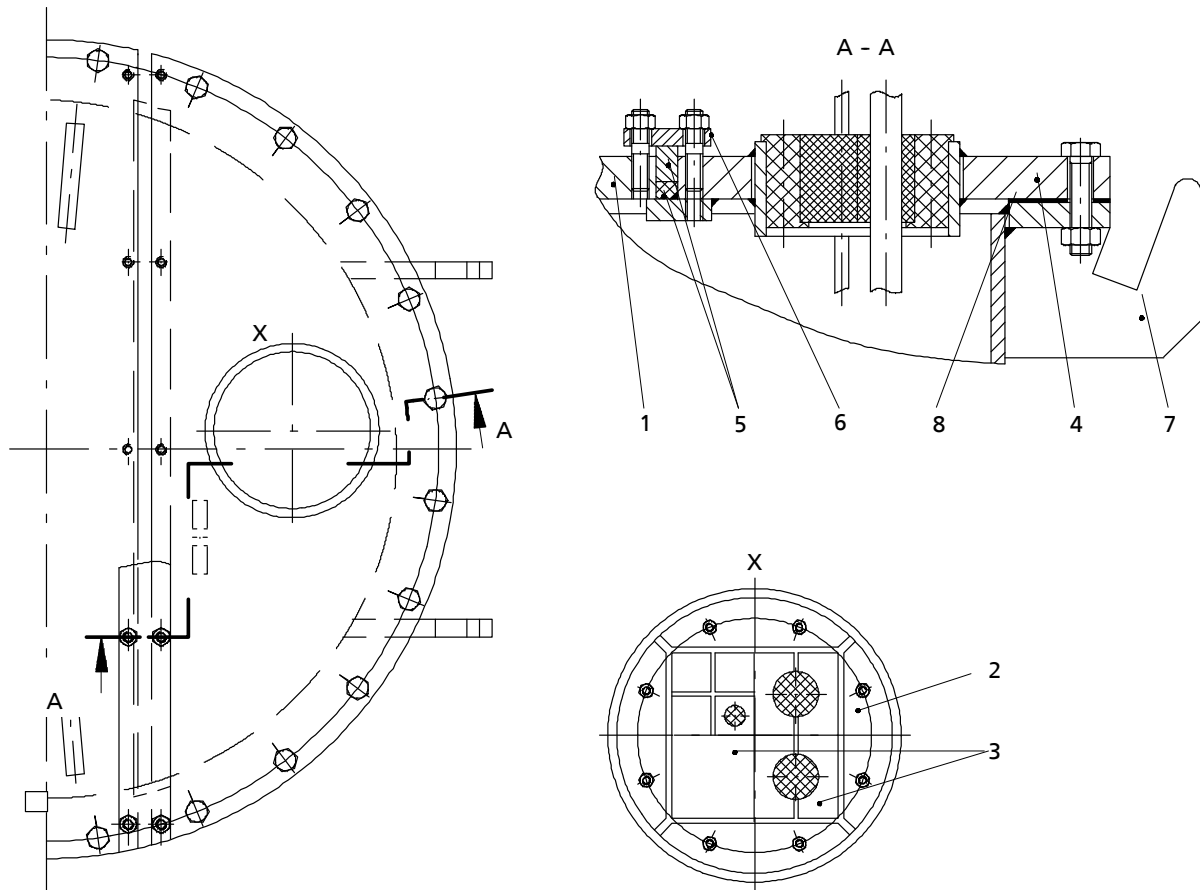


Fig. 4: Design variant with transit frame (up to 1 bar)

Table 39: List of components

Part No.	Description
1	Discharge tube cover ⁴⁸⁾
2	Transit frame (cable gland)
3	Packing blocks and insert blocks
4	Cover segment with cable gland
5	Closed-cell profile seal in groove between the two cover parts
6	Sealing arrangement of groove between the two cover parts
7	Support brackets for cover segment with cable glands
8	Gasket (e.g. rubber with fabric reinforcement)

⁴⁸ Discharge tube cover also available in single-piece design.

General assembly drawing with list of components

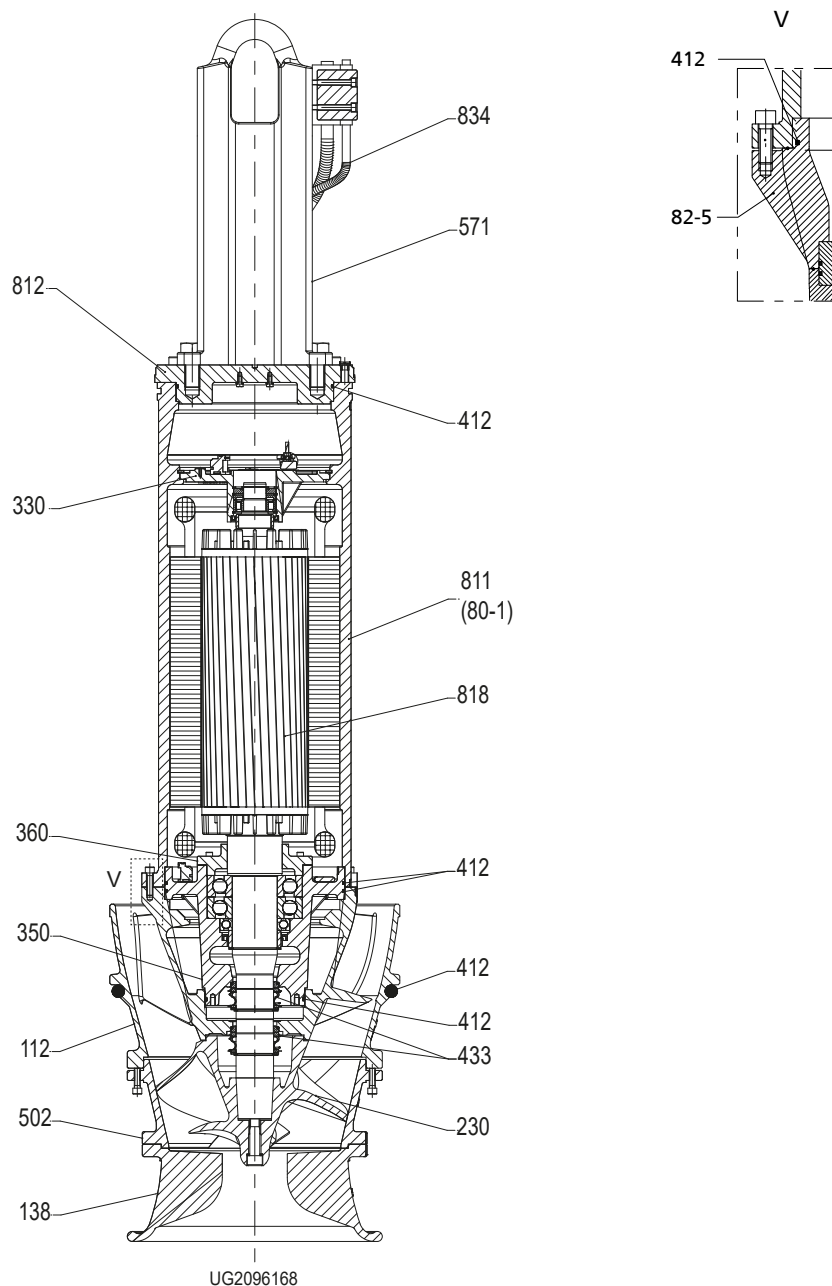


Table 40: List of components

Part No.	Description	Part No.	Description
80-1	Motor unit	412	O-ring
82-5	Adapter	433	Mechanical seal
112	Pump bowl	502	Casing wear ring
138	Bellmouth	571	Bail
230	Impeller	811	Motor housing
330	Bearing bracket	812	Motor housing cover
350	Bearing housing	818	Rotor
360	Bearing cover	834	Cable gland

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